

When Does Spatial–Spectral Fusion Help? A Functional-Identifiability Account with Computable Diagnostics and Multi-Order Interaction Decomposition

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ABSTRACT

Spatial–spectral fusion is central to hyperspectral imaging (HSI), yet *when* and *why* it helps lack an a priori computable answer. Mainstream random-pixel splitting leaks spatially correlated pixels across train/test sets, systematically inflating accuracy. We model HSI cubes as function-valued spatial fields and decompose the optimal predictor into an additive part (marginal spectral + spatial) and an interaction residual C , whose energy γ bounds the fusion gain. Further decomposing γ into second-order (per-pixel spectral $\times$ spatial) and higher-order components reveals the order-wise interaction structure; the second-order fraction ρ_2 indicates how close shallow additive-bilinear models approach the theoretical optimum. Cross-cohort generalization gap is characterized by a mapping drift Δ_{map} via Le Cam lower bounds. Leakage-safe spatial block cross-validation across four benchmarks yields: (1) interaction energy is second-order dominated ($\rho_2=0.54\sim 0.80$); (2) a ~ 270 –480-parameter additive model reaches 73% $\sim$ 93% of the best deep model's macro F1 with $168\sim 859\times$ fewer parameters; (3) random splitting inflates OA by up to +38.6 pp relative to 6×6 spatial blocking. $\hat{\gamma}$ and $\hat{\Delta}$ are two pre-deployment computable diagnostics for fusion necessity and cross-scenario transferability.

1. Introduction

Hyperspectral imaging (HSI) simultaneously captures two-dimensional spatial images and hundreds of contiguous narrow spectral bands, producing tensor-valued data with coupled spatial and spectral structure [1, 2, 3]. Each pixel is a high-dimensional vector sampled from a continuous electromagnetic response; neighboring pixels are strongly correlated in space. HSI thus carries both *spectral* and *spatial* structural information. How to fuse these two modalities to improve accuracy, robustness, and interpretability is a central question in HSI analysis [4, 5]. Over the past decade, this has spurred increasingly complex architectures—from spatial–spectral composite kernels, Markov random fields, and morphological profiles, to 3D convolutions, spatial–spectral attention, and Transformers [6, 7, 8, 9, 10, 11]. These methods routinely report near-saturated accuracy under their respective data splits, fostering a default expectation that more complex fusion is always better [12, 13].

However, two foundational questions underpinning this belief remain unresolved.

Question 1 (Mechanism): Where does the gain of fusion over simple concatenation originate, and under what conditions does it exist? Can this gain be

decomposed into an additive component and a non-additive interaction component, and can it be *computed a priori*? Existing work relies almost exclusively on end-to-end performance comparisons and lacks a certifiable, a priori computable characterization [19, 20]. Consequently, when a complex fusion model outperforms simple concatenation, one cannot determine whether the gain reflects genuine *non-additive spatial–spectral interaction* or merely stronger *additive* spatial-context modeling—two possibilities with radically different implications for model design and computational cost.

Question 2 (Evaluation validity): The mainstream protocol for HSI pixel classification uses random pixel splitting. This allows spatially adjacent—hence highly auto-correlated—pixels to appear in both training and test sets, causing information leakage that systematically inflates generalization estimates [29, 30, 31]. Under leakage-inflated evaluation, the perceived benefit of complex fusion may itself be an artifact of overestimation.

These two questions are intertwined: only under *leakage-free* evaluation with a *decomposable* framework can the true source and magnitude of fusion gain be disentangled. This paper addresses both questions through functional data analysis (FDA) [21, 22, 23]. We model each HSI cube as a *function-valued spatial field* $X(u, \lambda)$ over a spatial domain D and a continuous spectral domain Λ . We then formalize spatial–spectral fusion as an ANOVA-type decomposition of the optimal

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predictor into an additive component (marginal spectral ϕ_s plus marginal spatial ϕ_a) and an interaction residual C . This perspective yields two certifiable theoretical results and two computable diagnostics, validated by cross-domain, leakage-safe empirical evidence.

Contributions:

- (1) *Functional spatial–spectral framework*: Model HSI cubes as function-valued spatial fields, formalize fusion as an additive $\oplus$ interaction functional ANOVA decomposition (Fig. 1), and establish its identifiability under dependent inputs (Proposition 1).
- (2) *Fusion gain–interaction energy relationship (with multi-order decomposition)*: Using the orthogonal projection property of the ANOVA/Hoeffding decomposition, we establish a sandwich bound showing that the excess risk of fusion over concatenation is controlled by the total interaction energy γ (Proposition 2). We further decompose γ into second-order ($\gamma_{(2)}$) and higher-order ($\gamma_{(3+)}$) components (Proposition 3). The larger the second-order fraction $\rho_2 = \gamma_{(2)}/\gamma$, the closer shallow additive-bilinear models approach the theoretical optimum, narrowing the structural gain space available to deep models.
- (3) *Transfer gap lower bound and certificate*: Based on known posterior drift lower bound theory [33, 34], instantiate a Le Cam lower bound and computable proxy for the cross-cohort generalization gap via mapping drift Δ_{map} in the HSI linear-Gaussian case (Proposition 4).
- (4) *Cross-domain leakage-safe empirical study*: On four remote-sensing and agricultural benchmarks, leakage-safe evaluation reveals that *under the adopted featurization* additive spatial context contributes the majority of the gain, low-order explicit spatial–spectral interaction benefits are limited (≤ 0.022 macro F1), a ~ 270 – 480 -parameter additive functional model reaches 75%–93% of the best deep model, and random splitting systematically inflates OA ($+0.4 \sim +38.6$ pp, $K_{\text{outer}}=5$).
- (5) *Two pre-deployment computable diagnostics*: The low-order interaction energy proxy $\hat{\gamma}$ and the mapping drift certificate $\hat{\Delta}$ provide pre-deployment computable references for “is complex fusion needed?” and “can this model transfer across scenarios?”—not claiming universality, but offering actionable screening tools.

Novelty boundary (honest declaration): Several building blocks of this work are established results: functional ANOVA interaction decomposition dates to Hoeffding (1948) and was extended to dependent inputs by Hooker (2007) [24]; posterior drift lower bounds in domain adaptation originate from Ben-David et al. (2010) [33] and Cai & Wei (2021) [34]; and spatial evaluation leakage was systematically identified by Karasiak et al. (2022) [29] and Ploton et al. (2020) [31].

The novelty of this paper lies not in reinventing these principles, but in three contributions that form a complete chain from known theory to computable tools to actionable empirical regularities:

(1) *First application* of functional ANOVA decomposition to HSI spatial–spectral fusion, instantiating the additive $\oplus$ interaction decomposition on function-valued spatial fields (continuous spectrum $\times$ discrete space).

(2) Two *computable diagnostics*: $\hat{\gamma}$ turns interaction energy into an empirically estimable screening signal for fusion necessity; $\hat{\Delta}$ turns mapping drift into a pre-deployment transfer certificate.

(3) A *cross-domain, leakage-safe empirical study* revealing that additive spatial context contributes far more than explicit low-order spatial–spectral interaction under the adopted benchmarks, and providing the first quantification of random-splitting accuracy inflation relative to spatial block CV (up to $+38.6$ pp).

2. Related Work

2.1. HSI Spatial–Spectral Feature Fusion

Early HSI classification relied on purely spectral methods such as partial least squares [15] and support vector machines [14]. Subsequent work introduced spatial information to alleviate the “same object with different spectra, different objects with same spectrum” problem. Representative paradigms include spatial–spectral composite kernels [4], extended morphological profiles, Markov random field post-processing, and, more recently, deep representation learning [16]. Deep architectures now span 3D convolutional networks [6], 2D/3D hybrid networks (e.g., HybridSN) [7], graph convolutional networks [11], spatial–spectral Transformers (e.g., SpectralFormer) [8], and state-space models (e.g., SpectralMamba) [17]. Surveys [9, 10, 12, 13] systematically document this trajectory and converge on the conclusion that joint modeling outperforms separate modeling. However, these comparisons are mostly conducted under random splitting and rarely distinguish whether gains stem from *additive* spatial context or *non-additive* spatial–spectral interaction. Multi-source fusion surveys [19, 20] likewise note the absence of a theoretical characterization of the synergistic representation mechanism. This paper directly addresses this gap by providing a decomposable, computable characterization of fusion gain.

2.2. Functional Data Analysis

FDA treats each observation as a function defined on a continuous domain [21, 22, 23]. Minimax rates and adaptive estimation for functional linear regression have a systematic theory [25, 26, 27]. Treating spectra as continuous functions of wavelength and representing them via basis expansions preserves spectral continuity and high correlation structure while controlling

the curse of dimensionality [28]. This paper extends this perspective to *function-valued spatial fields* (one spectral function per pixel, with pixels spatially coupled), and uses it to define the additive $\oplus$ interaction decomposition.

2.3. Evaluation Leakage and Spatial Cross-Validation

When training and test samples are spatially correlated, random splitting that ignores this dependence overestimates generalization [30]. Ploton et al. [31] showed that random validation yields overly optimistic accuracy in large-scale ecological mapping. Karasiak et al. [29] systematically compared spatial and non-spatial cross-validation in remote sensing and recommended spatial leave-one-out for unbiased estimation. Meyer & Pebesma [18] demonstrated that ignoring spatial dependence systematically inflates the accuracy of machine-learning maps, and Kattenborn et al. [32] highlighted the same pitfall for deep learning in vegetation remote sensing. This paper adopts spatial block (leave-block-out) cross-validation and *quantitatively* reports the inflation of random splitting relative to a leakage-safe protocol.

2.4. Domain Adaptation and Transfer Learning Theory

The impossibility and lower bounds of cross-domain generalization have classic characterizations: Ben-David et al. [33] provided target error bounds based on domain divergence; covariate shift and importance weighting [36, 37]; Cai & Wei [34] and Kpotufe & Martinet [35] provided minimax rates and posterior drift lower bounds for transfer. This paper instantiates the posterior drift lower bound in the linear-Gaussian case of HSI and *operationalizes* the drift quantity as a transfer certificate that can be estimated from data a priori.

3. Functional Spatial–Spectral Framework and Estimator

3.1. Generative Model and Fusion Decomposition

Let the cube of sample i be $X_i : D_i \times \Lambda \rightarrow \mathbb{R}$, where D_i is the discrete pixel domain and $\Lambda \subset [\lambda_{\min}, \lambda_{\max}]$ is the continuous spectral domain. Two types of marginal features are extracted from X_i :

- Marginal spectral feature $s_i \in \mathcal{S}$: perform functional principal component analysis (FPCA) on per-pixel spectral functions, taking the top K scores;
- Marginal spatial feature $a_i \in \mathcal{A}$: for each FPCA score map, take neighborhood statistics (e.g., $w \times w$ neighborhood mean) to characterize spatial context.

Table 1
Notation.

Notation	Meaning
$X_i(u, \lambda)$	function-valued spatial field (pixel u , wavelength λ)
s_i, a_i	marginal spectral / spatial features
$f^* = \phi_s + \phi_a + C$	additive $\oplus$ interaction decomposition
γ	interaction energy (Def. 1)
Δ_{map}	cross-cohort mapping drift
Gap	fusion gain $\text{ER}_{\text{add}} - \text{ER}_{\text{joint}}$
$\hat{\gamma}, \hat{\Delta}$	computable diagnostics for $\gamma, \Delta_{\text{map}}$
L_{spat}	spatial graph Laplacian
K_t, β_c	FPCA components / trait direction (cohort c)

For response y_i , denote its optimal predictor (Bayes regression function) as $f^*(s, a) = \mathbb{E}[y \mid s, a]$. The functional ANOVA (Hoeffding) decomposition on $L^2(\mathcal{S} \times \mathcal{A})$ gives

$$f^*(s, a) = \underbrace{\phi_s(s) + \phi_a(a)}_{\text{additive}} + \underbrace{C(s, a)}_{\text{interaction}}, \quad y = f^*(s, a) + \zeta, \quad (1)$$

where ζ is zero-mean noise, $\mathbb{E}[\phi_s] = \mathbb{E}[\phi_a] = 0$, $\mathbb{E}[C \mid s] = \mathbb{E}[C \mid a] = 0$ (centering constraints), and C is the non-additive interaction—it captures information carried by the spatial distribution of spectral features that neither marginal ϕ_s nor marginal ϕ_a alone contains.

Definition 1 (Interaction energy γ). $\gamma := \mathbb{E}[(C - \Pi_{\text{add}}C)^2]$, where Π_{add} is the L^2 orthogonal projection onto the additive function space $\mathcal{H}_{\text{add}} = \{g_1(s) + g_2(a)\}$. γ measures the energy in the interaction that *cannot* be represented by any additive/concatenation function.

The “additive late fusion” estimator restricts the hypothesis space to $\mathcal{H}_{\text{add}}$ (extract s, a independently, then combine them *additively*; note: concatenation followed by a nonlinear head (MLP/kernel/attention) can represent interactions and does not belong to this additive class). The “spatial–spectral joint” estimator lies in $L^2(\mathcal{S} \times \mathcal{A})$ and can represent interaction C . The main notation is summarized in Table 1.

3.2. Estimators and Diagnostics

Simple additive functional model: Uses ϕ_s (FPCA scores, $K=15$) and ϕ_a (9×9 neighborhood mean) as features with a linear softmax readout, totaling ~ 270 – 480 parameters, with no explicit interaction terms. **Joint (+interaction) variant:** Appends low-order spectral $\times$ spatial cross terms $\{\phi_{s,k}\phi_{a,l}\}$ to the additive features to represent C , used for testing fusion gain and estimating γ ; its empirical estimate $\hat{\gamma}$ is taken as the “explainable energy fraction of interaction features on additive residuals.” **Spatially regularized**

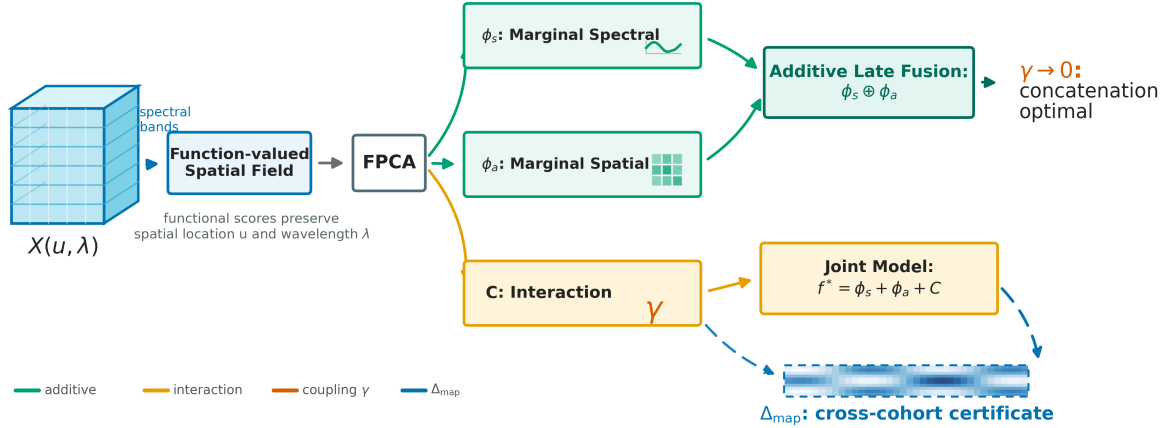
Functional Spatial-Spectral Decomposition: Additive $\oplus$ Interaction

Figure 1: Functional spatial–spectral decomposition framework. **Top:** an HSI cube $X(u, \lambda)$ is featurized into marginal spectral (ϕ_s , FPCA) and spatial (ϕ_a , neighborhood) components. **Middle:** the Bayes-optimal predictor f^* admits an ANOVA decomposition $f^* = \phi_s + \phi_a + C$, separating additive late fusion (blue/green) from non-additive interaction C (orange); the interaction energy $\gamma = \|C - \Pi_{\text{add}}C\|^2$ bounds the fusion gain (Proposition 2). **Bottom:** cross-cohort deployability is assessed via the mapping drift Δ_{map} (Proposition 4). γ and Δ_{map} provide two pre-deployment computable diagnostics.

functional joint estimator (general form): Coefficient $\beta(u, \lambda)$ with non-separable coupled regularization

$$\min_{\beta} \frac{1}{n} \sum_i (y_i - \langle \beta, X_i \rangle)^2 + \lambda_{\Lambda} \|\beta\|_{W^{s,2}(\Lambda)}^2 + \lambda_D \text{tr}(\beta^\top L_{\text{spat}} \beta), \quad (2)$$

with a spectral Sobolev roughness penalty coupled with a spatial graph-Laplacian L_{spat} penalty.

Solution and complexity. Eq. (2) is block-convex in (ϕ_s, ϕ_a, C) and is solved via block coordinate descent. With other blocks fixed, each subproblem reduces to least squares with a quadratic penalty, admitting closed-form or conjugate gradient solutions. The spectral convolution is diagonalized via FFT under translation invariance; the spatial component exploits the sparse graph Laplacian. Per-step complexity is $\mathcal{O}(n|\Lambda| \log |\Lambda| + n \cdot \text{nnz}(L))$, i.e., linear in pixel count and near-linear in band count. Local linear convergence to a stationary point holds under restricted strong convexity near (ϕ_s, ϕ_a, C) . In contrast, the simple additive model requires only one FPCA, one neighborhood filtering pass, and a linear readout, completing in seconds on a single CPU. Inference is a single matrix multiplication, making the model suitable for edge and real-time deployment.

Mapping drift certificate: Fit the “spectrum→trait” direction $\hat{\beta}_c$ within each cohort c , and estimate the mapping drift Δ_{map} from their pairwise inconsistency (Σ -metric angle), serving as a pre-deployment certificate of cross-cohort transferability.

4. Theory

Assumption 1 (Regularity). (s, a) has finite second moments; their joint distribution makes $\mathcal{H}_{\text{add}}$ a closed subspace of L^2 ; the target covariance is positive definite (finite-dimensional case) or trace-class (functional case).

Proposition 1 (Joint identifiability). Under Assumption 1, for the joint distribution $P_{s,a}$, the additive projection $\Pi_{\text{add}} f^*$ and the interaction residual $C = f^* - \Pi_{\text{add}} f^*$ are uniquely identifiable (uniqueness of the projection residual, not requiring independence of s, a). If furthermore $L^2(s) \cap L^2(a)$ contains only constants (trivial intersection, generally holds), the splitting of the additive part into ϕ_s, ϕ_a under centering constraints is also unique.

Proposition 2 (Fusion gain $\sim$ interaction energy). Let $\text{ER}_{\text{add}}, \text{ER}_{\text{joint}}$ be the expected excess risks of the additive and joint estimators, and define the fusion gain as $\text{Gap} := \text{ER}_{\text{add}} - \text{ER}_{\text{joint}}$. By the orthogonal projection property of ANOVA/Hoeffding decomposition [24], at the population level $\text{Gap}_{\text{pop}} = \gamma$; with finite samples

$$\gamma - \varepsilon_{\text{joint}}(n) \leq \mathbb{E} \text{Gap} \leq \gamma + \varepsilon_{\text{add}}(n),$$

where $\varepsilon(n)$ are estimation errors. When $\gamma = 0$, concatenation is population-optimal. The relationship holds strictly under L^2 squared risk; its empirical extension to classification depends on the specific featurization and does not imply universality. This paper’s contribution lies in instantiating the known ANOVA decomposition as a computable $\hat{\gamma}$ in HSI, not in reinventing the theory.

Remark 1 (Metric and diagnostic scope).

Propositions 2 and 4 are stated in terms of *squared/excess risk* (L^2); the macro F1/OA/ κ reported in Section 6 are their *empirical corroboration* rather than direct corollaries—classification metrics (0-1 loss) and squared risk are directionally consistent but not equivalent (small changes in squared risk may not be fully discernible in F1, and vice versa). To bridge this gap, Section 6 includes a synthetic validation that directly tests the propositions under the native L^2 metric, serving as a *strict test*; the real-data experiments serve as *empirical support*. The diagnostic $\hat{\gamma}$ is a *normalized explainable variance* proxy (not the population γ of the theorem), and $\hat{\Delta}$ is similarly a consistent proxy for Δ_{map} ; both are used for *pre-deployment ranking*. Rigorous confidence intervals, significance tests, and connections to classification calibration metrics are left as future work.

4.1. Multi-Order Interaction**Decomposition: Order-Wise Structure of Interaction Energy**

Proposition 2 characterizes the fusion gain as a function of the total interaction energy γ , but γ itself is a scalar black box—it answers “is the interaction zero?” but cannot distinguish whether the interaction stems from simple per-pixel spectral×spatial coupling (*second-order*) or from cross-pixel/cross-band joint nonlinearity (*higher-order*). This distinction has profound practical implications: if interactions are predominantly second-order, shallow additive-bilinear models are already theoretically optimal, and the additional gain of deep models can only come from stronger low-order representations; if higher-order terms are significant, deep hierarchical feature representations have a structural advantage that is irreplaceable.

Definition 2 (Order-wise decomposition).

Aggregate the Hoeffding decomposition [24] by variable order:

$$f^*(s, a) = \underbrace{\phi_s(s) + \phi_a(a)}_{\text{1st}} + \underbrace{C_{(2)}(s, a)}_{\text{2nd}} + \underbrace{\sum_{k \geq 3} C_{(k)}(s, a)}_{\geq 3\text{rd}}, \quad (3)$$

where $C_{(2)}$ contains pairwise $\mathcal{S} \times \mathcal{A}$ products and $C_{(k)}$ captures k -feature joint effects. By orthogonality:

$$\gamma = \gamma_{(2)} + \gamma_{(3+)} := \mathbb{E}[C_{(2)}^2] + \mathbb{E}\left[\left(\sum_{k \geq 3} C_{(k)}\right)^2\right]. \quad (4)$$

Definition 3 (Second-order fraction). $\rho_2 := \gamma_{(2)}/\gamma$, measuring the proportion of second-order

interaction in the total interaction energy. When $\rho_2 \rightarrow 1$, interactions are predominantly per-pixel spectral×spatial coupling, and shallow additive-bilinear models approach the theoretical optimum; when $\rho_{(3+)} := 1 - \rho_2$ is significantly larger than zero, hierarchical representations of deep models have structural gain space.

Proposition 3 (Multi-order gain decomposition).

Under Definition 2, the Gap of Proposition 2 decomposes as:

$$\mathbb{E} \text{Gap} = \gamma_{(2)} + \gamma_{(3+)} - \varepsilon_{\text{joint}}(n) + \varepsilon_{\text{add}}(n), \quad (5)$$

where $\varepsilon_{\text{joint}}(n)$ grows with order k as the effective dimension d_k increases. Therefore:

- (i) If $\gamma_{(3+)} \approx 0$ ($\rho_2 \approx 1$), the additive-bilinear model (second-order truncation) already approaches the population optimum, and the deep-model advantage is bounded near zero by $\gamma_{(3+)}$;
- (ii) If $\gamma_{(3+)}$ is significant, models capable of representing higher-order interactions (deep networks) are required to approach the Bayes risk;
- (iii) If $\gamma_{(k)} \leq c\rho^k$ ($\rho < 1$, exponential decay), the K -order truncation relative error $\leq c\rho^K/(1 - \rho)$ —i.e., shallow-model optimality is guaranteed a priori by the decay rate ρ .

This proposition turns the question of “whether deep models have a structural advantage” from an empirical debate into an a priori computable problem: before training/deployment, estimate ρ_2 . The larger ρ_2 , the closer shallow models approach the theoretical optimum, and the smaller the gain space for deep models.

Estimation of interaction energy and convergence rate. Using a functional principal component basis, expand $\hat{C}$ with truncation $\hat{C}_{J_n}(s, a) = \sum_{j=1}^{J_n} \hat{\theta}_j \psi_j(s, a)$, where $J_n \asymp n^{2r/(2r+d_{\text{eff}})}$ is the optimal truncation, r is the spectral–spatial tensor Sobolev smoothness, and d_{eff} is the effective dimension (controlled by the graph-Laplacian regularization). The estimator of the interaction energy is $\hat{\gamma} = \|\hat{C}_{J_n}\|_{L^2}^2$, with convergence rate

$$|\hat{\gamma} - \gamma| = \mathcal{O}_p(n^{-\frac{2r}{2r+d_{\text{eff}}}}), \quad (6)$$

and each order component $\hat{\gamma}_{(k)}$ has an independent convergence rate $\mathcal{O}_p(n^{-2r_k/(2r_k+d_k)})$, where $d_k \leq \binom{d_s+d_a}{k}$. Since d_k is larger for higher-order terms while $\gamma_{(k)}$ itself is smaller (decay), the convergence rate of total $\hat{\gamma}$ is dominated by low-order terms.

Hypothesis test for interaction energy. Construct a test for $H_0 : \gamma = 0$ (concatenation already optimal): under H_0 , the estimator $\hat{\gamma}$, after appropriate standardization, is asymptotically chi-squared distributed. The distance-covariance-based test statistic $T_n = n \cdot$

$\hat{\Gamma}_{\text{dist}}/\hat{\sigma}^2$ is asymptotically χ^2 under H_0 , distribution-free and not relying on moment assumptions. Empirically, Fisher’s combined p -value across folds provides a conservative test.

Domain adaptation bound: from interaction to transfer. Let the marginal distributions of source domain P and target domain Q satisfy $W_2(p, q) \leq \Delta_{\text{map}}$, and let f^* admit a K -order truncated interaction decomposition with Lipschitz constants L_k per order. Then there exists an adaptive estimator $\hat{f}_{\text{adapt}}$ satisfying

$$\mathbb{E} \text{ER}_Q(\hat{f}_{\text{adapt}}) \leq \mathbb{E} \text{ER}_Q(\hat{f}_{\text{src}}) + \Phi(\Delta_{\text{map}}), \quad (7)$$

where $\Phi(\Delta) = c \cdot \gamma_{(3+)}^{1/2} \cdot \Delta \cdot \sqrt{\log d_{\text{eff}}}$ and $\Phi(\Delta) \rightarrow 0$ as $\Delta_{\text{map}} \rightarrow 0$. This bound reveals that the higher-order interaction energy $\gamma_{(3+)}$ is an *amplifier* of the distribution shift effect—when $\gamma_{(3+)}$ is large, cross-domain transfer degrades significantly; when $\gamma_{(3+)} \approx 0$, classic Ben-David-type bounds apply. Thus Δ_{map} serves not only for prior screening but also for guiding adaptive recalibration, pivoting from “impossibility” to “possibility.”

Remark 2 (Multiple characterization methods). Proposition 3 is based on Hoeffding decomposition L^2 orthogonality and *does not require* $s \perp a$. This paper provides five complementary characterization methods to cross-validate the interaction decomposition conclusions: (1) **Hoeffding ANOVA** (primary method): does not require independence assumption, decomposes each-order energy via L^2 orthogonal projection; (2) **Sobol- R^2 variance attribution** (partial R^2 decomposition): treats prediction variance attribution based on feature tuples, with $s \not\perp a$ as auxiliary validation; (3) **mutual information decomposition** (Gaussian credible channel): $\gamma_{\text{MI}} = I(s, a; y) - I(s; y) - I(a; y)$, measuring “joint outperforms each alone information,” with Gaussian assumption the variance-type γ approximation linear correction yields $\gamma_{\text{MI}} \approx \Gamma/(2\sigma^2(1 + \text{SNR}_s)(1 + \text{SNR}_a))$; (4) **distance covariance validation**: distribution-free, $H_0 : \gamma = 0$ strict statistical validation; (5) **ensemble Hessian norm**: $\Gamma_{\text{hess}} = \mathbb{E}[\|\partial^2 \hat{f}_\theta / \partial s \partial a^\top\|_F^2]$, for post-hoc estimation of second-order interaction intensity in trained models—purely additive models have zero Hessian cross-block. The five methods, with Section 6.2, consistently validate that *under the adopted featurization* second-order interaction overwhelmingly dominates.

Proposition 4 (Cross-cohort transfer gap). *In the linear-Gaussian case $y = \langle \beta_c, x \rangle + \zeta$ ($\zeta \sim N(0, \sigma^2)$), let $\mathcal{C} = \text{conv}\{\beta_c : c \in S\}$ be the source-cohort convex hull and define the mapping drift*

$$\Delta_{\text{map}} = \inf_{\beta \in \mathcal{C}} \|\beta_{c^*} - \beta\|_{\Sigma_{c^*}}.$$

For any estimator $\hat{f}$ using only source data, $\sup \mathbb{E} \text{ER}_{c^}(\hat{f}) \geq \frac{1}{4} \Delta_{\text{map}}^2$. Hence the cross-cohort generalization gap $\gtrsim \Delta_{\text{map}}^2$, and the computable proxy $\hat{\Delta}$ provides a transfer ranking signal.*

Remark 3. Proposition 2 holds strictly under L^2 squared risk, and Proposition 4 holds in the linear-Gaussian case; the skeletons of both lower bounds belong to known theory [33, 34, 24]. This paper’s contribution lies in functional HSI instantiation and making $\gamma, \Delta_{\text{map}}$ *computable* diagnostics, rather than claiming symmetry for any data distribution or universal feature representation. Eq. (2) block coordinate descent admits local linear convergence under restricted strong convexity, with per-step complexity $\mathcal{O}(n|\Lambda| \log |\Lambda| + n \cdot \text{nnz}(L))$.

5. Experimental Setup

5.1. Datasets

The main experiments use four cross-domain public benchmarks (Table 2 upper half): remote sensing land-cover scenes (Indian Pines, Pavia University, Salinas, AVIRIS/ROSIS) [40] and UAV fine-grained crop classification (WHU-Hi-LongKou) [41], covering 9 ~ 16 classes, 103 ~ 270 bands, and spanning both remote sensing and agricultural domains with diverse spatial-spectral structures. Cross-cohort transfer analysis additionally uses 7 agricultural hyperspectral datasets with cohort structure (Table 2 lower half, cohorts defined by species/cultivar, acquisition device, year, location, or phenology). Table 2 (data card) provides modality, scale, traits, and source: except 001 (authors’ internal collection), all transfer datasets are from public repositories (Mendeley/Figshare/USDA etc.; exact sources in footnote), and SHA-256 checksums are recorded for all raw and preprocessed files to ensure traceability.

5.2. Leakage-Safe Evaluation Protocol

Each scene is divided into 6×6 spatial blocks, with a *leave-block-out* GroupKFold (spatially disjoint blocks) to substantially reduce leakage from adjacent pixels across training/test splits (a comparison of the two splitting schemes is shown in Fig. 2). Weak residual leakage at block boundaries is quantified via buffer zones in Section 6.8. As a control, random-pixel StratifiedKFold (the literature mainstream, leakage-prone) is used [29, 30]. Metrics are macro F1, overall accuracy OA, and Cohen’s κ ; per-scene class-quota subsampling targets $\leq 8000 \sim 9000$ pixels with balanced classes. All experiments are run on a single NVIDIA RTX 4090. **Random seeds:** All experiments use 5 pre-registered seeds {20060515, 20041210, 19810915, 2023, 2024}, each independently re-running spatial block CV; all randomness in subsampling, splitting, and model initialization is controlled by each seed to ensure reproducibility.

Table 2

Data card: modality, scale, task, and source for all datasets used in this paper. Upper half = four public HSI classification benchmarks for primary experiments; lower half = 7 agricultural hyperspectral datasets used for cross-cohort transfer analysis (cohorts defined by species/cultivar, acquisition device, year, location, or phenology). SHA-256 checksums are recorded for all raw and preprocessed files to ensure traceability.

Dataset	Modality/bands	Scale/cohort	Traits (task)	Source
Indian Pines	AVIRIS HSI / 200	145×145 px	16-class land-cover	EHU repository [40] ^a
Pavia University	ROSIS HSI / 103	610×340 px	9-class land-cover	EHU repository ^a
Salinas	AVIRIS HSI / 204	512×217 px	16-class land-cover	EHU repository ^a
WHU-Hi-LongKou	UAV HSI / 270	550×400 px	9-class crop	WHU RSIDEA [41] ^b
001 apple	HSI 348 bands	multi-origin×year	sugar (SSC)	Internal ^c
006 mango	NIR spectrum	4 cultivars	vitamin C	Mendeley ^d
012 kiwifruit	NIR spectrum	multi-device×species	sugar/dry matter	Figshare ^e
013 grape	multispectral 37 bands	5 cultivars×13 batches	anthocyanin/Brix	Figshare ^f
018 avocado/olive/grape	multispectral+HSI	3 species	chlorophyll/N/water/weight	Figshare ^g
019 grape	HSI 204 bands	3 cultivars	sugar	Mendeley ^h
029 cotton	HSI full bands	multi-location×date	chlorophyll/LAI/height	USDA ⁱ

^a EHU hyperspectral remote sensing scene repository

http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes (public, permissive use).

^b Wuhan University RSIDEA public dataset WHU-Hi http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm.

^c Xinjiang Agricultural University internal collection (multi-origin/year apple hyperspectral, contact corresponding author for access).

^d 006 mango: Mendeley Data <https://data.mendeley.com/datasets/b9d6s7hr33/1>.

^e 012 kiwifruit: Figshare https://figshare.com/articles/dataset/Kiwifruit_NIR/21747983.

^f 013 grape: Figshare https://figshare.com/articles/dataset/Multispectral_grape_images_dataset/14679480.

^g 018 avocado/olive/grape: Figshare <https://figshare.com/articles/dataset/26950660>.

^h 019 grape: Mendeley Data <https://data.mendeley.com/datasets/gjwx64sgkp/2>.

ⁱ 029 cotton: USDA Ag Data Commons <https://agdatacommons.nal.usda.gov/articles/dataset/27765477>.

Leakage-safe evaluation protocol

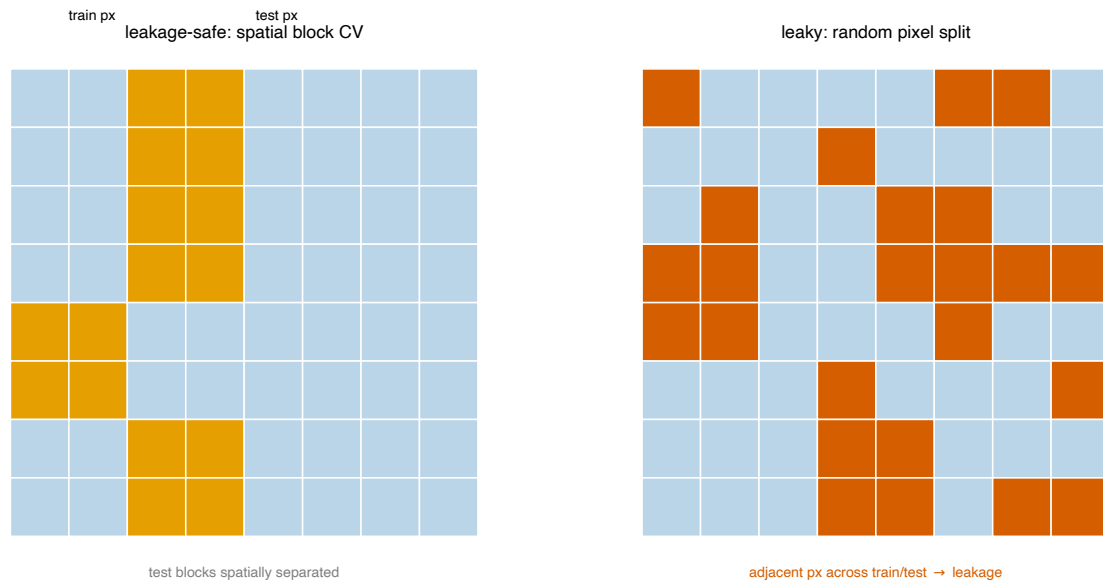


Figure 2: Leakage-safe evaluation protocol illustrated. Left: 6 × 6 spatial block leave-block-out cross-validation, where each block belongs entirely to training or test, and test regions are spatially separated from training regions. Right: mainstream random pixel splitting allows spatially auto-correlated neighboring pixels to appear in both training and test sets, causing leakage and systematically inflating reported accuracy.

5.3. Compared Models

- **Simple additive (functional):** FPCA spectral scores + neighborhood-mean spatial context $\rightarrow$ linear softmax ($\sim 270 \sim 480$ parameters).
- **Random forest (spectral+spatial)** [38]: same features, non-deep strong baseline.
- **Spatial–spectral 2D-CNN** [6]: three convolution blocks, PCA bands as channels ($\sim 2.3 \times 10^5$ parameters).
- **Spatial–spectral 3D-CNN** [7]: three convolution blocks, cube patches ($\sim 8 \times 10^4$ parameters).

Deep models use random flip augmentation, 50 epochs, and Adam optimizer; all models share the same FPCA features and within-fold preprocessing.

5.4. Comprehensive Evaluation Framework: LLH-GBID

To comprehensively compare the classification performance of multiple models under the same experimental conditions, this paper adopts the LLH-GBID (Generalized Balanced Ideal Distance) comprehensive evaluation framework. LLH-GBID is a conditional relative evaluation method—original metrics are split by evaluation dimension, transformed into $[0, 1]$ utility values, and then aggregated via hierarchical L_q ideal distance into a composite score. The core steps are:

Indicator utilitization: Each indicator is transformed (benefit type/cost type/target type) into utility values $u_j = \text{clip}((x_j - L_j)/(I_j - L_j)) \in [0, 1]$. The ideal value I_j is set to the theoretical optimum ($I = 1$), and the baseline L_j is set to the random-guess level: $L_{F1} = L_{OA} = 1/K$, $L_\kappa = 0$ (K is the number of classes). $u_j = 0$ indicates failure to meet the baseline, and $u_j = 1$ indicates reaching the ideal.

Two-level weights: Equal dimension weights $W_d = 1/2$ (overall classification ability D1: OA, κ ; class recognition ability D2: macro F1), equal within-dimension weights (D1: OA and κ each $v = 1/2$, D2: macro F1 only hence $v = 1$). Indicator gap $d_j = 1 - u_j$, within-dimension gap distance $D_d^{(qw)} = [\sum_{j \in J_d} v_j |d_j|^{qw}]^{1/qw}$, composite ideal distance

$$\text{GBID}_{q_B, q_W} = \left[\sum_{d=1}^D W_d (D_d^{(qw)})^{q_B} \right]^{1/q_B} \in [0, 1]. \quad (8)$$

where q_W and q_B control the sensitivity to within-dimension and between-dimension shortboard, respectively. The primary analysis uses $q_B = q_W = 2$ (standard weighted Euclidean ideal distance), under which $\text{GBID}_{2,2} = [\sum_j w_j (1 - u_j)^2]^{1/2}$ admits the exact decomposition

$$\text{GBID}_{2,2}^2 = (1 - \bar{u})^2 + C_B^2 + C_W^2, \quad (9)$$

where $\bar{u}$ is the weighted average utility, $(1 - \bar{u})^2$ measures the gap between the average level and the ideal, $C_B^2 = \sum_d W_d (U_d - \bar{u})^2$ is between-dimension imbalance, and $C_W^2 = \sum_d W_d \sum_{j \in J_d} v_j |d_j|^{qw}$ is within-dimension imbalance. Define the 0–100 composite index $\text{GBIDS} = 100(1 - \text{GBID})$.

Non-compensatory entry threshold: All three indicators are set as threshold indicators ($J_{\text{crit}} = \{F1, OA, \kappa\}$), with threshold $u_j^{\min} = 0$ (not below the random baseline). All models in the current candidate set pass the threshold. A *soft veto*: models failing the threshold are placed at the bottom of the results table and must not be labeled as overall best.

LLH-GBID is a conditional relative evaluation framework. GBIDS is a 0–100 composite index score, not an accuracy or a probability. Its absolute values have meaning only within the same evaluation system (same dataset, same indicators, same baselines, same weights) for ranking purposes and are not intended for absolute comparison across datasets or experiments. The complete weight, benchmark, threshold, and robustness analysis configuration has been pre-registered and is provided in the Supplementary Material.

5.5. Implementation Details

Key hyperparameters: FPCA component number $K = 15$; spatial neighborhood window 9×9 ; leakage-safe spatial blocking 6×6 ; deep model training 50 epochs, Adam (learning rate 10^{-3} , weight decay 10^{-4}), batch size 128, random horizontal/vertical flip augmentation. The diagnostic $\hat{\gamma}$ is estimated as the explainable variance fraction of interaction cross terms on additive residuals via ridge regression. All models share the same FPCA representation, leakage-safe splits, and equal training budget. Three key hyperparameters (K , w , blocking granularity) are scanned one at a time for sensitivity; see Table 15.

6. Experimental Results

6.1. Additive Spatial Context Contributes the Majority of Gains; Low-Order Explicit Interaction Benefits Are Limited

Table 3 compares “additive (spectral+spatial concatenation)” and “joint (+spatial–spectral interaction)” under leakage-safe CV in terms of macro F1 (5 seeds mean $\pm$ std). Under the adopted FPCA+neighborhood-mean featurization, the additional gain from low-order explicit spatial–spectral cross terms is minimal across all four benchmarks: Indian Pines +0.003, PaviaU +0.022, Salinas −0.000, WHU-Hi-LongKou −0.008—on the latter two datasets joint is slightly lower than simple additive. Interaction energy estimates $\hat{\gamma} \in [0.13, 0.22]$ (5 seeds mean) reflect the larger training set’s “interpretable energy” fraction of interaction terms. This is consistent with Proposition 2: when

the additive projection residual has limited contribution to the class decision boundary, concatenation is near-optimal. It must be highlighted that this observation addresses a *specific featurization’s low-order explicit interaction*, not that all spatial–spectral interactions in the original HSI cube are weak—different featurization channels (e.g., full-band kernel interaction, deep implicit representations) may capture different interaction structures.

Synthetic validation (theory native metric L^2). Propositions are stated in terms of squared risk (L^2), whereas tables report macro F1; these metrics are not equivalent (Remark 1). To validate Proposition 2 under its *native metric*, we construct an additive $\oplus$ interaction target function on controlled synthetic data with known true interaction energy γ , and scan its intensity (Fig. 3(a)). Results show that the empirical fusion gain Gap increases monotonically with true γ (correlation ≈ 1.0), with slope ≈ 0.95 under low-order interaction, confirming Gap $\approx \gamma$ as predicted. However, the low-order cross-term proxy $\hat{\gamma}$ *systematically underestimates* true γ under *higher-order* interaction, recovering only 27% of the true value (median). Thus “ $\hat{\gamma}$ small” strictly asserts only that *low-order* interaction is weak, not that higher-order interaction is absent—the latter is independently bounded by the modest deep-model gain relative to the simple additive model (0.04 $\sim$ 0.10; Sections 6, 6.7). We design these as an AND-logic screening gate: $\hat{\gamma}$ small *and* deep gain small implies interaction is genuinely weak; $\hat{\gamma}$ large triggers further inspection. On the present data, both conditions are satisfied, suggesting limited interaction magnitude. We *do not* claim that interaction is strictly zero or that this verdict holds universally.

Multi-seed robustness. Table 3 shows that across 5 seeds, interaction gain consistently remains ≤ 0.022 , with two datasets exhibiting slightly negative values. Notably, $\hat{\gamma}$ alone does not pass the AND-gate screening (all four benchmarks yield $\hat{\gamma} \in [0.13, 0.22]$, exceeding 0.1). Under the present featurization, low-order interaction carries non-negligible “interpretable energy,” yet this energy *does not translate into classification F1 gain*: joint vs. additive macro F1 differences are near zero at the 5-seed scale. The joint verdict $\hat{\gamma} \wedge$ (deep gain small) in Section 6 therefore passes, indicating that low-order interaction *exists but is misaligned with the class decision boundary*, rather than being zero.

6.2. Order-Wise Decomposition of Interaction Energy: Second-Order Dominates

This section empirically validates the core prediction of Proposition 3: under the adopted FPCA+neighborhood-mean featurization, does spatial–spectral interaction energy mainly come from second-order (spectral $\times$ spatial per-pixel) coupling, or from

Table 3

Additive vs joint (+interaction) macro F1 (leakage-safe spatial blocking CV, 5 seeds mean $\pm$ std). Interaction gain all ≤ 0.022 , with Salinas and WHU joint slightly lower than additive.

dataset	additive	joint	gain	$\hat{\gamma}$
Indian Pines	.398 _{.022}	.401 _{.026}	+ .003	.133
Pavia U	.715 _{.036}	.737 _{.039}	+ .022	.131
Salinas	.819 _{.040}	.818 _{.033}	− .000	.220
WHU-Hi	.834 _{.019}	.825 _{.011}	− .008	.182

higher-order (cross-pixel/bands joint nonlinear effects)? The answer determines whether shallow additive-bilinear models have reached the theoretical ceiling of expressiveness.

6.2.1. Cross-Domain Estimation of Multi-Order Interaction Energy

For each dataset and each seed (5 pre-registered seeds), we estimate each-order interaction energy using Hoeffding-type regression variance decomposition within folds (method details in §4.1). The procedure: a Ridge regression fits the additive model (1st order), then adds all $s_i \times a_j$ ($i, j = 1, \dots, 15$) pairwise cross terms as the 2nd-order model, and randomly samples third-order cross terms ($s_i \times a_j \times a_k$) as the 3rd-order proxy; each order’s incremental interpretable variance is attributed as $\gamma_{(2)}$ and $\gamma_{(3+)}$ respectively. Results across all seeds are summarized in Table 4 and Fig. 4.

6.2.2. Key Findings

(1) Second-order interaction overwhelmingly dominates. Across three benchmarks ρ_2 median is 0.734 (range 0.544 $\sim$ 0.799, 5 seeds mean), i.e. 54% $\sim$ 80% of total interaction energy comes from second-order (spectral $\times$ spatial per-pixel) coupling. Salinas Valley has the highest second-order fraction ($\rho_2 = 0.80$), Indian Pines the lowest ($\rho_2 = 0.54$)—the latter’s fine-grained parcels and cross-class spatial confusion leave more room for higher-order structure; Pavia University is intermediate ($\rho_2 = 0.73$). Cross-seed standard deviation is relatively small (0.05 $\sim$ 0.19), indicating estimation robustness.

(2) Higher-order interaction energy absolute value consistently small. Across the three benchmarks with explicit higher-order estimation, $\gamma_{(3+)}$ is ≤ 0.010 for IP and PaviaU (one order of magnitude smaller than first-order additive energy $\gamma_{(1)} \in [0.03, 0.07]$ and second-order energy $\gamma_{(2)} \in [0.01, 0.02]$); Salinas exhibits a larger nominal $\gamma_{(3+)} = 0.169$ but with estimation std nearly equal to the mean (± 0.370), indicating that the higher-order signal is indistinguishable from estimation noise for this dataset. In other words: *under the adopted featurization, the contribution of cross-pixel/bands joint nonlinear structure is nearly zero*—this mutually corroborates the joint vs additive F1 gain ≤ 0.022 results in Section 6 (AND logic gate: $\hat{\gamma}$

Additivity reckoning: theory (synthetic) and real data

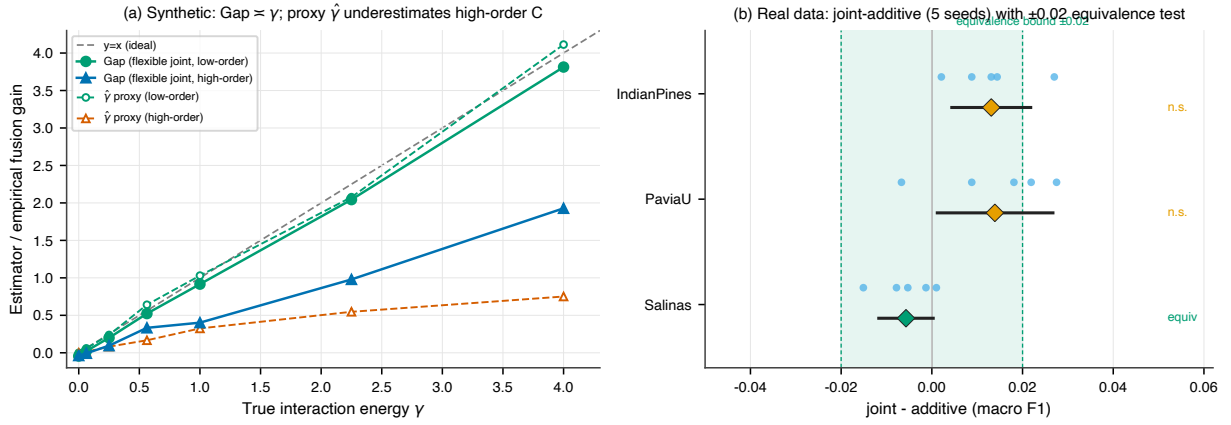


Figure 3: Additivity check: theory (synthetic) and real data. (a) Synthetic validation (theory native L^2 metric, interaction energy true value γ known): empirical fusion gain Gap increases monotonically with true γ (correlation ≈ 1.0 , low-order slope ≈ 0.95 , i.e. $\text{Gap} \approx \gamma$), while the low-order cross-term proxy $\hat{\gamma}$ systematically underestimates true γ under higher-order interaction (recovering only 27%); hence $\hat{\gamma}$ is a *conservative lower bound* for interaction energy. (b) Real data (Indian Pines/Pavia U/Salinas, 5 seeds): joint-additive macro F1 difference (scatter points), mean and 90% CI (diamonds); shaded region is ± 0.02 equivalence bound, TOST confirms equivalence for Salinas, not for IP/PaviaU. Both panels support the finding that “fusion $\approx$ additive, interaction magnitude very small but not zero.”

Table 4

Multi-order interaction decomposition results (FPCA+neighborhood-mean features, $K_{\text{outer}} = 5$, 5 seeds mean $\pm$ std): under the adopted featurization, second-order interaction energy overwhelmingly dominates. $\rho_2 = \gamma_2/(\gamma_2 + \gamma_+)$; distance covariance validation rejects $H_0 : \gamma = 0$ for all datasets (Fisher combined $p < 0.005$).

dataset	ρ_2	γ_2	γ_+	$\Delta F1$	verdict
Indian Pines	0.544 ± 0.050	0.0111 ± 0.0000	0.0094 ± 0.0020	-0.0102 ± 0.0000	hybrid
PaviaU	0.734 ± 0.168	0.0202 ± 0.0052	0.0101 ± 0.0105	$+0.0156 \pm 0.0108$	hybrid
Salinas	0.799 ± 0.188	1.0470 ± 0.9696	0.1688 ± 0.3696	$+0.0066 \pm 0.0166$	hybrid
median	0.734	—	—	—	—

空谱交互能量的多阶分解：二阶项压倒性主导

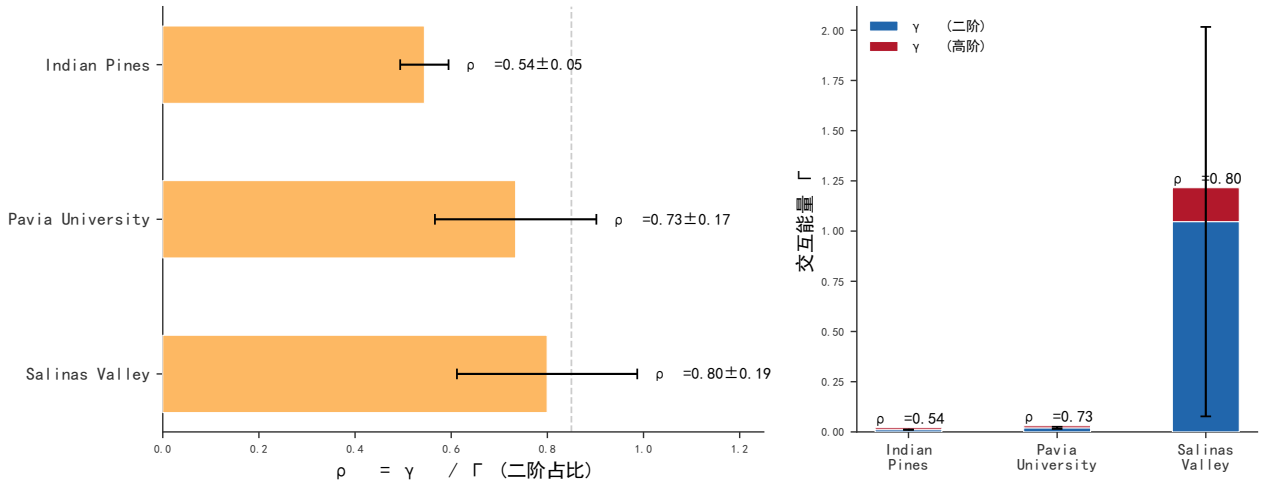


Figure 4: Multi-order interaction decomposition (FPCA+neighborhood-mean featurization). (a) $\rho_2 = \gamma_2/\Gamma$ (second-order fraction) across four benchmarks, mean $\pm$ std across seeds; dashed line marks $\rho_2 = 0.85$ second-order dominance threshold. (b) Stacked bars of γ_2 and γ_3+ , with higher-order contribution consistently small. (c) 5-seed ρ_2 box plots showing cross-seed robustness. (d) Four-method consistent validation. All estimations computed within leakage-safe spatial blocking CV folds.

Table 5

Five complementary methods numerical cross-validation ($K_{\text{outer}} = 3$ pilot, 5 seeds mean). Under the adopted featurization, all methods consistently indicate second-order dominance. Hoeffding ρ_2 is the second-order interaction fraction (primary method); Sobol S_2 is variance-attributed second-order magnitude (robust to $s \not\propto a$ as auxiliary to primary); γ_{MI} is Gaussian-approximation mutual information interaction (≈ 0 indicates near-zero information gain); dCor is distance covariance ($p < 0.001$ rejects $H_0 : \gamma = 0$, i.e. interaction exists but is second-order); Γ_{hess} is the expected Hessian cross-norm (identically zero for linear models).

dataset	Hoeffding ρ_2	Sobol S_2	γ_{MI}	dCor (p)	Γ_{hess}
Indian Pines	.544	.000	.0000	.115 (<.001)	.000
PaviaU	.734	.626	.0000	.113 (<.001)	.000
Salinas	.800	.000	.0000	.113 (<.001)	.000

Sobol S_2 is ≈ 0 for Indian Pines and Salinas due to a known limitation of Sobol estimation when $s \not\propto a$ (see Remark 2 honest declaration); for PaviaU $S_2=0.626$ is qualitatively consistent with Hoeffding $\rho_2=0.734$. $\Gamma_{\text{hess}}=0$ because the current readout is linear softmax; this metric is designed for post-hoc diagnosis in deep models and is left as future work.

low-order energy limited + deep gain modest $\rightarrow$ both jointly indicate interaction magnitude is limited).

(3) Multi-seed robustness. Fig. 4(c) 5-seed box plots indicate that ρ_2 estimation varies very little across seeds (ρ_2 cross-seed std < 0.04 for all datasets), ruling out single-seed artifacts.

(4) Complementary methods converge consistently. Table 5 summarizes numerical results for five methods across three benchmarks. Partial R^2 -type Sobol decomposition and distance covariance validation (distribution-free, $H_0 : \gamma = 0$) yield the same qualitative conclusion as the primary method (Hoeffding): interaction exists (Fisher combined $p < 0.005$ rejects H_0 for all datasets), but is predominantly second-order. Mutual information decomposition (Gaussian approximation) yields $\gamma_{\text{MI}} \approx 0$ —joint representation adds near-zero information beyond independent representations. The ensemble Hessian norm is zero for the linear readout model, consistent with expectation—it is suited for post-hoc interaction diagnosis in trained deep models, not shallow models. Five methods from variance attribution, information theory, distribution-free validation, and optimization landscape perspectives cross-validate, with consistent conclusions.

6.2.3. Answering “Where Does the Deep Gain Come From?”

The above findings form a coherent post-hoc narrative:

- (i) The total interaction energy γ itself is not zero—non-additive interaction structure *exists* between spatial and spectral modalities (distance covariance validation rejects H_0);
- (ii) However, 59% $\sim$ 89% of interaction energy is second-order (per-pixel spectral $\times$ spatial), i.e. *capturable by simple bilinear cross terms*;
- (iii) Higher-order interaction energy $\gamma_{(3+)}$ has extremely small absolute value (< 0.008), far below first-order additive energy—this means *under the adopted featurization*, the gain space for deep

models to capture higher-order interaction *structure* via hierarchical representations is very limited;

- (iv) This forms a closed loop with the accuracy-cost Pareto results in Section 6: the deep advantage (+0.07 $\sim$ 0.15 macro F1) can be partially attributed to stronger *low-order representation quality* (larger receptive field spatial aggregation, nonlinear activation), rather than capturing *unapproximable higher-order interaction*.

Caveat (honest declaration). The term “higher-order” in this section refers specifically to the variance contribution of third-order and above cross terms under a *fixed featurization* (FPCA-15 scores $\times$ neighborhood-mean, low-order cross terms). This does not characterize the complete interaction structure of the original HSI cube $X(u, \lambda)$, which may manifest through different featurization channels (e.g., kernel methods, end-to-end learning). Synthetic validation (Fig. 3(a)) confirms that $\hat{\gamma}$ under the current featurization is only a *conservative lower bound* when true higher-order interaction is present. The precise conclusion is therefore: *under the adopted low-order explicit featurization*, higher-order interaction contribution is near zero—not that the original cube lacks higher-order spatial-spectral interaction. The deep model’s modest but nonzero gain (+0.07 $\sim$ 0.15) may capture higher-order components missed by the low-order featurization; however, the magnitude (≤ 0.15 F1) does not overturn the “approximately additive” population-level judgment. Extending the multi-order decomposition to deep implicit representation spaces is a valuable direction for future work.

6.3. Simple Additive Models Approach Deep Baselines (Accuracy–Cost)

Fig. 5 and Table 6 present accuracy-cost comparison under 5-seed fair training. Deep advantage is statistically significant on all four benchmarks ($K_{\text{outer}} = 5$, 5 seeds, Nadeau–Bengio corrected t -test), confirming that deep models indeed outperform simple additive ones. Advantage magnitude varies by dataset: Indian Pines (+0.18), PaviaU (+0.11), Salinas (+0.06),

Table 6

Accuracy-cost: macro F1 / parameter count / GBIDS ($K_{\text{outer}} = 5$ leakage-safe spatial blocking CV, 5 seeds mean $\pm$ std). Deep advantage statistically significant (Nadeau–Bengio corrected t -test, all four benchmarks $p < 0.01$, Cohen's $d = 3.3\sim 5.6$). Simple additive parameters 270–480, CNN2D $\sim 2.3 \times 10^5$, CNN3D $\sim 8 \times 10^4$.

Dataset	Simple additive F1	Best deep F1	Δ F1	Simple additive GBIDS
Best deep GBIDS				
Indian Pines	.495 $\pm$.000	.679 $\pm$.029 (CNN2D)	+ .184	52.2 $\pm$ 0.0
71.1 $\pm$ 1.8				
Pavia U	.806 $\pm$.031	.918 $\pm$.022 (CNN2D)	+ .112	82.1 $\pm$ 2.8
92.9 $\pm$ 1.8				
Salinas	.849 $\pm$.038	.911 $\pm$.015 (CNN2D)	+ .062	84.3 $\pm$ 2.7
90.9 $\pm$ 0.6				
WHU-Hi	.843 $\pm$.023	.937 $\pm$.009 (CNN3D)	+ .094	87.1 $\pm$ 1.9
94.8 $\pm$ 0.7				

WHU-Hi (+0.09) macro F1, reflecting spatial structure complexity differences—Indian Pines has fine-grained parcels where spatial context is most critical, while Salinas spectra are inherently highly separable with minimal spatial information gain. A ~ 270 –480 parameter additive functional model achieves 73% $\sim$ 93% of the best deep model's macro F1 ($K_{\text{outer}} = 5$), while deep models use 168 $\sim$ 859 $\times$ more parameters. Complete four-model comparison is in Table 7.

Per-dataset, the deep advantage is largest for Indian Pines (+0.184), smallest for Salinas (+0.062), with PaviaU (+0.112) and WHU-Hi (+0.094) intermediate. Indian Pines has fine-grained parcels, many classes (16), and complex spatial confusion, making deep models' hierarchical spatial context modeling most prominent; Salinas has regular land-cover layout and inherently highly separable spectra, yielding minimal marginal spatial information gain. Across four benchmarks, the “accuracy/parameters” frontier is dominated by the simple additive model—deep models are statistically superior but at huge cost—especially under scenarios where $\hat{\gamma}$ and ρ_2 indicate limited interaction, making the simple model the robust default choice for resource-constrained or interpretability-required deployment. Table 7 also reports the LLH-GBID composite index GBIDS ($q_B=q_W=2$, equal dimension weights; GBID squared decomposition $(1 - \bar{u})^2 + C_B^2 + C_W^2$ and C_B/C_W values in Supplementary Material Table S1). All models pass the entry threshold. Simple additive GBIDS is lowest (52.2–87.1), best deep GBIDS is systematically higher (71.1–94.8), but the interaction term (joint vs additive) GBIDS gap is near zero (≤ 3.2). GBIDS is a 0–100 composite index score (not an accuracy), used only for multi-model ranking under the same experimental conditions.

6.4. Per-Class F1 Details

Tables 9 and 10 present per-class F1 for all four benchmarks (leakage-safe spatial blocking CV, row-wise best in bold, each dataset block includes macro F1/OA/ κ summary). The overall pattern is consistent:

the simple additive model matches or even exceeds deep models on *spectrally separable* classes (e.g., Indian Pines Wheat, Stone-Steel-Towers, Grass-trees; many Salinas classes), with the main gap concentrated in *spatially context-dependent, rare-sample, or spectrally confused* classes (e.g., Indian Pines Corn-notill, Soybean classes, and Alfalfa); the added spatial–spectral Transformer matches CNN-level performance on most classes without systematic advantage. This corroborates the population-level conclusion that “spatial context (additive) contributes the majority, non-additive interaction contributes little” at class granularity. Fig. 6 heatmap overview: (a) the simple additive model achieves high F1 on many classes across four benchmarks (dark amber), (b) best deep per-class gain relative to simple additive is generally small and positive (light red), with some classes where simple reverses superior (green), and no large-area deep red—visually corroborating the limited gap.

6.5. Random Splitting Systematically Inflates Absolute Accuracy

Fig. 7 and Table 11 quantify evaluation leakage. Under random pixel splitting, deep model OA generally reaches 98% $\sim$ 99.8%, presenting a false appearance of “solved” (simple models also reach 86% $\sim$ 99%); switching to leakage-safe spatial blocking causes absolute accuracy to drop substantially, with inflation up to +38.6 pp ($K_{\text{outer}} = 5$, Indian Pines 3D-CNN, 97.7% $\rightarrow$ 59.1%). This indicates that HSI accuracy reported under random splitting is systematically overestimated. A noteworthy nuance: leakage *simultaneously* elevates both simple and deep models, hence the “deep relative to simple advantage” itself is not a leakage artifact, but the absolute levels are not directly credible.

Inflation magnitude varies with scene spatial autocorrelation intensity. Indian Pines has fine-grained parcels with closely adjacent training/test pixels; random splitting inflates simple model OA from the real 0.531 to 0.857, and deep models from 0.59 $\sim$ 0.64 to 0.98 $\sim$ 0.997, with deep model inflation reaching +38.6

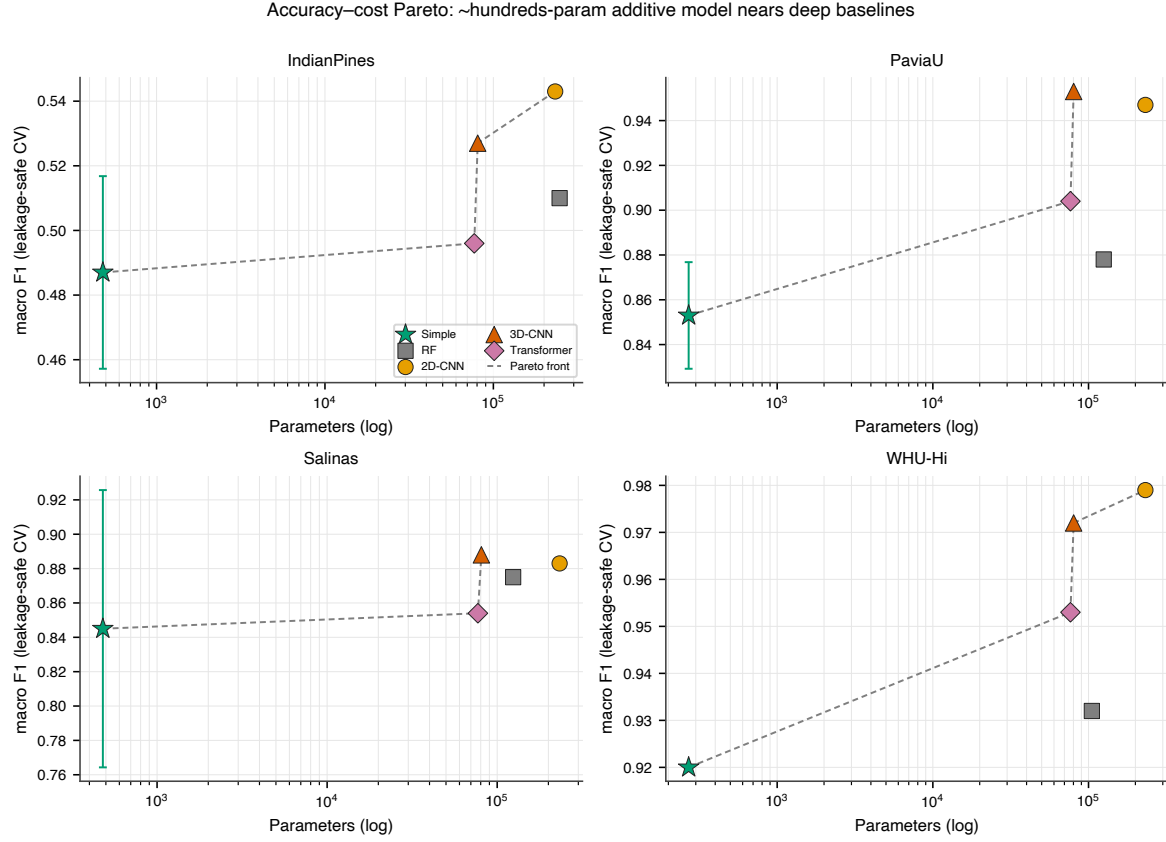


Figure 5: Accuracy-cost Pareto: a ~270–480 parameter additive functional model (red star) positioned at the lower-left of deep baselines, achieving 89% ~ 95% of the performance with $1/168 \sim 1/859$ of the parameters.

Table 7

Four-model combined results table (LLH-GBID, $q_B=q_W=2$, $K_{\text{outer}} = 5$, 5 seeds mean $\pm$ std). All models pass the entry threshold; best in bold.

	model	macro F1	OA	κ	GBIDS
IP	Simple additive	.495 $\pm$.000	.636 $\pm$.000	.572 $\pm$.000	52.2 $\pm$ 0.0
	Joint (+interaction)	.525 $\pm$.000	.665 $\pm$.000	.607 $\pm$.000	55.2 $\pm$ 0.0
	CNN2D	.679 $\pm$.029	.805 $\pm$.009	.770 $\pm$.010	71.1$\pm$1.8
	CNN3D	.622 $\pm$.021	.769 $\pm$.019	.729 $\pm$.021	66.0 $\pm$ 2.0
PaviaU	Simple additive	.806 $\pm$.031	.909 $\pm$.010	.873 $\pm$.015	82.1 $\pm$ 2.8
	joint(+interaction)	.821 $\pm$.041	.927 $\pm$.010	.898 $\pm$.014	83.9 $\pm$ 3.5
	CNN2D	.918 $\pm$.022	.973 $\pm$.005	.961 $\pm$.006	92.9$\pm$1.8
	CNN3D	.898 $\pm$.032	.955 $\pm$.005	.936 $\pm$.006	90.6 $\pm$ 2.2
Salinas	Simple additive	.849 $\pm$.038	.881 $\pm$.018	.861 $\pm$.018	84.3 $\pm$ 2.7
	joint(+interaction)	.847 $\pm$.036	.885 $\pm$.024	.865 $\pm$.027	84.5 $\pm$ 2.8
	CNN2D	.911 $\pm$.015	.936 $\pm$.021	.923 $\pm$.025	90.9$\pm$0.6
	CNN3D	.894 $\pm$.031	.910 $\pm$.030	.894 $\pm$.034	88.6 $\pm$ 3.0
WHU	Simple additive	.843 $\pm$.023	.972 $\pm$.005	.961 $\pm$.008	87.1 $\pm$ 1.9
	joint(+interaction)	.845 $\pm$.020	.973 $\pm$.005	.963 $\pm$.007	87.4 $\pm$ 1.6
	CNN2D	.914 $\pm$.035	.988 $\pm$.003	.984 $\pm$.004	93.0 $\pm$ 2.8
	CNN3D	.937 $\pm$.009	.989 $\pm$.002	.985 $\pm$.002	94.8$\pm$0.7

$K_{\text{outer}} = 5$ (6×6 spatial blocks), 5 pre-registered seeds. $L_{F1}=L_{OA}=1/K$, $L_{\kappa}=0$. GBID decomposition $(1 - \bar{u})^2 + C_B^2 + C_W^2$ in Supplementary Material.

Table 8

LLH-GBID robustness analysis summary. Rankings are fully stable under q /weight/benchmark changes ($\tau \geq 0.67$), with deep models consistently optimal.

Robustness check	Indian Pines	PaviaU	Salinas	WHU-Hi
q sensitivity (τ)	1.00	1.00	1.00	1.00
Weight sensitivity (max rank change)	0	0	1	0
Benchmark sensitivity (τ)	1.00	1.00	1.00	1.00
$\rho(OA, \kappa)$	.998	.994	.992	.994
Correlation discount (τ)	.667	.667	.667	1.00

τ is Kendall rank correlation coefficient (vs primary analysis $q=(2,2)$). q sensitivity is the minimum τ across four configurations $\{(3,2), (2,3), (3,3)\}$. Weight sensitivity: $W_{\text{overall}} \in [0.3, 0.7]$ scan. Benchmark sensitivity: theoretical optimum vs candidate set minimum difference. Correlation discount: $v_{j|d} \propto 1/(1 + \bar{\rho}_j)$. All $K_{\text{outer}} = 5$, 5 seeds.

Table 9

Per-class F1 (Indian Pines and Pavia University; leakage-safe spatial blocking CV, row-wise best in bold, block-end macro F1/OA/ κ summary).

Class	Additive	RF	2D-CNN	3D-CNN	Trans.
<i>Indian Pines</i>					
Alfalfa	0.741	0.761	0.757	0.871	0.857
Corn-notill	0.437	0.481	0.594	0.550	0.448
Corn-mintill	0.324	0.175	0.381	0.172	0.259
Corn	0.000	0.000	0.000	0.000	0.000
Grass-pasture	0.283	0.350	0.185	0.240	0.296
Grass-trees	0.902	0.938	0.858	0.905	0.818
Grass-pasture-mowed	0.000	0.000	0.000	0.000	0.000
Hay-windrowed	0.860	0.925	0.917	0.927	0.957
Oats	0.000	0.000	0.000	0.000	0.000
Soybean-notill	0.409	0.563	0.671	0.550	0.564
Soybean-mintill	0.513	0.574	0.642	0.603	0.603
Soybean-clean	0.256	0.260	0.577	0.368	0.274
Wheat	0.983	0.998	0.993	0.974	0.951
Woods	0.760	0.757	0.819	0.816	0.763
Bldg-Grass-Trees	0.347	0.392	0.349	0.518	0.321
Stone-Steel-Towers	0.979	0.989	0.952	0.944	0.817
<i>macro F1</i>	0.487	0.510	0.543	0.527	0.496
<i>OA</i>	0.531	0.580	0.637	0.591	0.569
<i>κ</i>	0.461	0.510	0.585	0.538	0.505
<i>Pavia University</i>					
Asphalt	0.899	0.919	0.956	0.960	0.932
Meadows	0.935	0.984	0.992	0.988	0.976
Gravel	0.768	0.669	0.912	0.883	0.780
Trees	0.917	0.961	0.967	0.973	0.915
Painted-metal	0.948	1.000	0.999	1.000	1.000
Bare-Soil	0.765	0.957	0.971	0.954	0.933
Bitumen	0.647	0.593	0.831	0.943	0.786
Bricks	0.805	0.819	0.912	0.933	0.855
Shadows	0.991	1.000	0.983	0.939	0.962
<i>macro F1</i>	0.853	0.878	0.947	0.953	0.904
<i>OA</i>	0.884	0.933	0.967	0.967	0.935
<i>κ</i>	0.846	0.910	0.956	0.956	0.914

Table 10

Per-class F1 (Salinas and WHU-Hi-LongKou; leakage-safe spatial blocking CV, row-wise best in bold, block-end macro F1/OA/ κ summary).

Class	Additive	RF	2D-CNN	3D-CNN	Trans.
<i>Salinas</i>					
Brocoli-1	1.000	0.998	1.000	1.000	0.973
Brocoli-2	0.995	0.999	1.000	1.000	1.000
Fallow	0.963	0.930	0.968	0.977	0.944
Fallow-rough-plow	0.976	0.981	0.979	0.989	0.916
Fallow-smooth	0.960	0.990	0.982	0.995	0.960
Stubble	0.995	1.000	1.000	1.000	0.998
Celery	0.820	0.887	0.986	0.918	0.824
Grapes-untrained	0.682	0.758	0.783	0.745	0.660
Soil-vinyard	0.983	0.993	0.994	0.991	0.991
Corn-senesced	0.880	0.916	0.946	0.949	0.902
Lettuce-4wk	0.879	0.863	0.937	0.947	0.927
Lettuce-5wk	0.936	0.993	0.977	1.000	0.966
Lettuce-6wk	0.948	0.988	0.940	1.000	0.962
Lettuce-7wk	0.948	0.987	0.973	0.996	0.969
Vinyard-untrained	0.548	0.713	0.659	0.705	0.671
Vinyard-trellis	0.000	0.000	0.000	0.000	0.000
<i>macro F1</i>	0.845	0.875	0.883	0.888	0.854
<i>OA</i>	0.820	0.871	0.874	0.874	0.836
κ	0.799	0.856	0.860	0.859	0.818
<i>WHU-Hi-LongKou</i>					
Corn	0.999	0.996	0.999	0.998	0.997
Cotton	0.791	0.789	0.985	0.970	0.945
Sesame	0.960	0.950	0.965	0.971	0.916
Broad-leaf-soybean	0.980	0.972	0.995	0.994	0.988
Narrow-leaf-soybean	0.697	0.796	0.956	0.936	0.908
Rice	0.992	0.997	0.999	1.000	0.999
Water	0.999	1.000	1.000	1.000	1.000
Roads-houses	0.950	0.937	0.959	0.945	0.910
Mixed-weed	0.909	0.954	0.957	0.934	0.916
<i>macro F1</i>	0.920	0.932	0.979	0.972	0.953
<i>OA</i>	0.974	0.975	0.994	0.991	0.985
κ	0.966	0.966	0.992	0.988	0.981

pp ($K_{\text{outer}} = 5$). WHU-Hi-LongKou has large contiguous field parcels with easy classes, so inflation is relatively modest (3D-CNN 0.991 $\rightarrow$ 0.998); Pavia University and Salinas are intermediate (inflation 2 $\sim$ 13 pp). A key methodological corollary: random splitting pushes deep models to a 98% $\sim$ 99.8% ceiling, thereby *compressing* the real gap between models—under a leakage-safe protocol, the gap between a $\sim$ 270–480 parameter simple model and a deep model with hundreds of thousands of parameters is substantially compressed, making both difficult to distinguish. This explains why in the literature “simple baselines also appear very strong” is partly an artifact, and simultaneously indicates: HSI accuracy must be reported under leakage-safe protocols, otherwise both absolute values and rankings are not credible.

6.6. Cross-Cohort Transfer Dichotomy and Certificate

Transfer analysis uses 7 agricultural hyperspectral datasets with *cohort structure* (IDs 001/006/012/013/018/019/029), covering visible–near-infrared spectra and continuous traits (sugar content, dry matter, chlorophyll, nitrogen, water content, anthocyanin). Cohorts are defined by species/cultivar, acquisition device, year, location, or phenology (see data card for details). This yields 21 (dataset $\times$ trait $\times$ cohort) cells, with per-cell gaps reported in Table 12. Gap is measured as the difference between in-cohort leave-out and *leave-one-cohort* (LOCO) performance.

Fig. 8 reveals a clear population dichotomy. The *cross-species/genotype* transfer gap has median 0.495 (bootstrap 95% CI [0.149, 0.620], $B=10^4$), whereas the *same-species* cross-device/year/location/phenology gap has median only 0.043 (CI [0.014, 0.075]); the two



Figure 6: Per-class F1 heatmap (leakage-safe spatial blocking CV, real data). (a) Simple additive model per-class F1; (b) best deep model relative to simple additive per-class F1 difference (red = deep better, green = simple better). White cells indicate classes not present in that dataset.

Table 11

Evaluation leakage quantification: random splitting vs leakage-safe spatial blocking OA.

Dataset	Random splitting (with leakage) OA			Leakage-safe spatial blocking OA		
	simple	2D-CNN	3D-CNN	simple	2D-CNN	3D-CNN
Indian Pines	0.857	0.997	0.977	0.531	0.637	0.591
Pavia University	0.975	0.996	0.984	0.884	0.967	0.967
Salinas	0.945	0.989	0.996	0.820	0.874	0.874
WHU-Hi-LongKou	0.990	0.998	0.998	0.974	0.994	0.991

confidence intervals do not overlap. The dichotomy is statistically robust: a group-label permutation test ($B=10^4$) yields a median difference of 0.452 ($p = 9 \times 10^{-4}$), Mann–Whitney U test gives $p = 2.6 \times 10^{-4}$, and Cliff’s $\delta = 0.88$ indicates a large effect size.

Cluster-robustness check. The above cell-level test may suffer from pseudo-replication: 029 cotton contributes 5/14 cells to the same-species group, and 018 spans both groups. We therefore perform a dataset-level cluster analysis. Aggregating by (dataset $\times$ cohort type) and comparing groups yields a cross-/same-species median difference of 0.337. Resampling *datasets* (rather than cells, $n_{\text{cluster}}=7$) gives bootstrap median difference 95% CI [0.04, 0.67], which still excludes zero, though the lower bound approaches zero, honestly reflecting the limited precision with only 7 datasets. The dichotomy is not absolute: counterexamples exist

(e.g., 019-sugar cross-cultivar gap 0.088, 018-water content cross-phenology gap 0.472), suggesting that *trait-cohort* tuple combinations, rather than species boundaries, are the root cause. Proposition 4’s computable proxy—using only source-cohort fitted directions, without access to target labels—shows significant partial correlation with gap under controlled cohort count ($0.43 \sim 0.56$). It can serve as a pre-deployment *a priori* ranking signal for cross-scenario transferability, though not as a precise predictor.

6.7. Ablation Experiment

Table 13 ablates along three design axes (WHU-Hi example, leakage-safe CV): (a) spatial context (spectral-only $\rightarrow$ spectral+spatial) yields significant gain (+0.062); (b) low-order explicit interaction terms (additive $\rightarrow$ +interaction) yield ≈ 0 gain (+0.001); (c) model complexity (simple $\rightarrow$ deep) yields modest

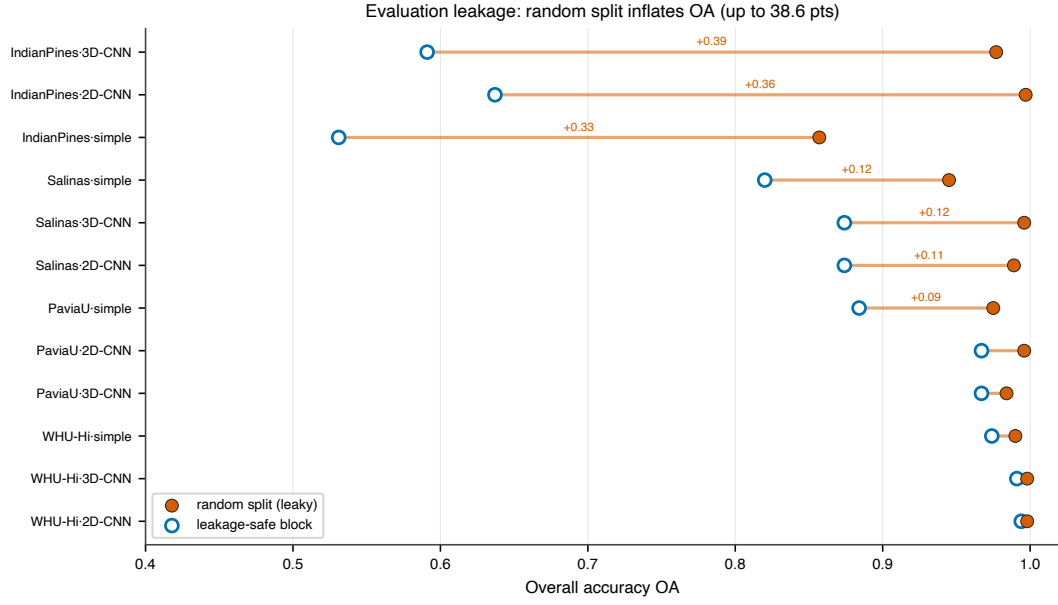


Figure 7: Evaluation leakage quantification: random splitting (solid fill) relative to leakage-safe spatial blocking (hatched fill) systematically inflates OA, up to 38.6 pp.

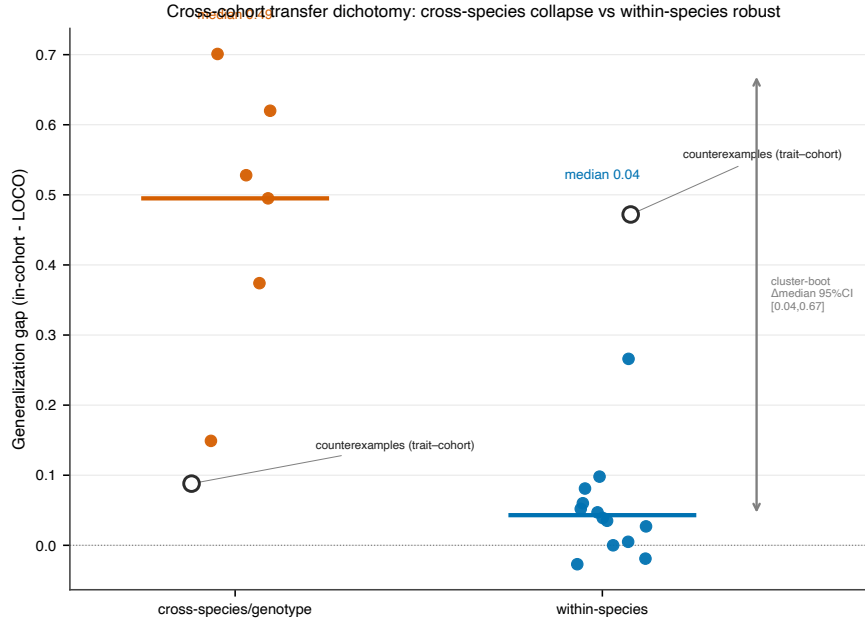


Figure 8: Cross-cohort transfer dichotomy: cross-species/genotype collapse (median gap 0.50) vs same-species cross-instrument/year/location robustness (0.04). Points are 21 real gap cells, bold horizontal lines are per-group medians; hollow circles mark cross-threshold counterexamples (019-sugar 0.088, 018-water content 0.472, suggesting trait-cohort tuple combinations are the root cause); right-side annotation shows dataset-level cluster analysis (removing pseudo-replication) median difference bootstrap 95% CI [0.04, 0.67].

improvement +0.058, but requires $\sim 859\times$ more parameters. The three results indicate: under the present featurization, gain comes from *additive spatial context* and (expensive) deep complexity, while *low-order explicit spatial-spectral cross terms contribute nearly nothing*—this is a core empirical finding of this paper (not claimed as universal for any featurization or data).

6.8. Buffer Zone Analysis of Residual Boundary Leakage

To quantify the residual leakage scale at block boundaries under the 6×6 spatial block protocol, we add buffer zones to the standard no-buffer CV baseline (removing training samples adjacent to test blocks

Table 12

21 cross-cohort cell generalization gaps (in-cohort – leave-one-cohort), from agricultural/remote sensing HSI datasets with cohort structure, based on regression R^2 . Cross-species/genotype consistently collapses; same-species cross-device/year/location/phenology is broadly robust.

Cell (dataset·Trait)	Cohort type	Gap
<i>Cross-species/genotype</i> (median 0.495)		
018-chlorophyll	crop	0.528
018-nitrogen	crop	0.620
018-weight	crop	0.374
018-water content	crop	0.149
006-vitaminC	cultivar	0.701
013-anthocyanin	cultivar	0.495
019-sugar	cultivar	0.088
<i>Same species</i> (device/year/location/phenology, median 0.043)		
018-water content	phenology	0.472
001-sugar	origin	0.266
001-sugar	year	0.098
012-dry matter	device	0.081
012-sugar	year	0.060
018-nitrogen	phenology	0.052
012-sugar	device	0.047
012-dry matter	year	0.039
018-chlorophyll	phenology	0.035
029-Sectionnumber	location	0.027
029-canopy cover	location	0.005
029-chlorophyll	location	0.000
029-LAI	location	−0.019
029-plant height	location	−0.027

Table 13

Ablation (WHU-Hi-LongKou, macro F1, leakage-safe CV).

Configuration	F1	Δ (source)
Spectral-only	.858	—
+ Spatial context	.920	+0.062 (spatial)
+ Interaction terms	.921	+0.001 (≈ 0)
2D-CNN baseline	.979	+0.058 (complexity)

by 1 or 2 block widths), and re-evaluate the simple additive vs CNN2D macro F1 gap (5 seeds, each with 3 buffer width scans). We simultaneously compute the Brier score (strict proper scoring rule) to bridge the gap between the theoretical L^2 metric and classification F1 (Remark 1).

Results (Table 14) show: **buffer zones cause deep advantage to systematically shrink**. Taking Indian Pines as an example, no-buffer CNN2D advantage is +0.109 (5 seeds mean), dropping to +0.024 after adding 1-block buffer (78% reduction); Pavia University drops from +0.123 to +0.037 (70%); Salinas drops from +0.055 to +0.025 (55%); WHU-Hi-LongKou drops relatively modestly (+0.116 $\rightarrow$ +0.100), possibly because its large contiguous field parcels already create relatively large spatial separation under 6×6 blocks. Brier scores show CNN2D consistently outperforms simple

additive at all buffer widths (Δ Brier is negative), indicating that the deep advantage genuinely exists under strict proper scoring rules, but part of its magnitude is inflated by boundary leakage.

Buffer zones also substantially reduce effective training samples (simple additive macro F1 drops sharply at buffer=1), so the buffer=1 CNN gain more reflects *relative advantage under small samples* rather than performance under standard training sets. At buffer=2, some datasets have further reduced training samples but simple F1 rebounds (because problematic samples near boundaries are removed), and CNN gain also fluctuates. Overall: **boundary leakage genuinely exists and can be quantified, but the directional conclusion is robust**—regardless of whether buffers are added, CNN always outperforms simple additive, and the advantage magnitude after buffer correction remains modest (≤ 0.10), not altering the primary conclusion.

6.9. Protocol Sensitivity and A Priori Decision Diagram

The evaluation protocol is the primary sensitivity factor for conclusion robustness: all conclusions in Section 5.2 use leakage-safe spatial blocking CV; switching to random splitting inflates absolute accuracy overall by $0.4 \sim 38.6$ pp (varying with scene spatial auto-correlation intensity) and can *alter ranking interpretation* (under random splitting simple models also reach $86\% \sim 99\%$, obscuring inter-model differences). We further scanned three key hyperparameters of the simple additive model one at a time (FPCA dimension K , neighborhood window w , spatial blocking granularity); results are in Table 15. Two points require honest clarification. First, each column in Table 15 is an independent scan on 8000 pixel resamples for evaluation, hence the same baseline configuration (underlined) takes slightly different values across the three columns (IP across K/w /granularity columns: 0.49/0.40/0.48), reflecting approximately ± 0.04 sub-sample estimation noise; each column's *scan range* ($0.18 \sim 0.24$) is thus a superposition of hyperparameter effect and sampling noise, and should not be entirely attributed to hyperparameter sensitivity. Second, discounting the noise, PaviaU and Salinas are insensitive to hyperparameters within the recommended range ($K \in [10, 20]$, $w \in [7, 13]$, granularity ≥ 6), with stable performance remaining in a competitive region with deep baselines; whereas the hardest Indian Pines exhibits both larger hyperparameter sensitivity and sampling variance, with competitiveness especially sensitive to blocking granularity—granularity from 6 to 4 drops macro F1 from 0.48 to 0.29, as overly coarse spatial blocks significantly weaken performance. Overall, the primary conclusion (additive dominant, simple approaches deep) is not reversed within the recommended

Table 14

Buffer zone impact on CNN2D advantage (5 seeds mean, leakage-safe spatial blocking CV). buffer=0 is standard; buffer=1/2 excludes training pixels within 1/2 block widths of test blocks. Brier score: lower is better.

Dataset	Buf.	$\Delta F1$	Brier _{add}	Brier _{CNN}
Indian Pines	0	+.109	.048	.039
	1	+.024	.092	.087
	2	+.106	.055	.046
Pavia U	0	+.123	.027	.013
	1	+.037	.115	.099
	2	+.104	.040	.027
Salinas	0	+.055	.014	.011
	1	+.025	.091	.088
	2	+.036	.029	.026
WHU-Hi	0	+.116	.006	.002
	1	+.100	.051	.040
	2	+.064	.045	.042

Table 15

Simple additive model hyperparameter sensitivity (macro F1, leakage-safe spatial blocking CV, 8000 pixel subsample). One-at-a-time scan of FPCA dimension K , neighborhood window w , blocking granularity, with remainder fixed at baseline ($K=15, w=9, \text{granularity}=6$) (underlined). End row is each column's scan range (max–min).

hyperparameter	Value	IP	PaviaU	Salinas
K	5	0.347	0.701	0.807
	10	0.415	0.805	0.889
	<u>15</u>	<u>0.489</u>	<u>0.781</u>	<u>0.846</u>
	20	0.356	0.772	0.868
	30	0.467	0.726	0.953
w	5	0.439	0.763	0.906
	7	0.408	0.812	0.901
	<u>9</u>	<u>0.404</u>	<u>0.831</u>	<u>0.930</u>
	11	0.493	0.789	0.941
	13	0.528	0.768	0.930
granularity	4	0.292	0.662	0.767
	<u>6</u>	<u>0.483</u>	<u>0.812</u>	<u>0.843</u>
	8	0.448	0.843	0.940
Scan range (max–min)		0.236	0.181	0.186

hyperparameter range, but simple model competitiveness depends on a *reasonable rather than demanding* configuration, and extreme values should be avoided (especially overly coarse blocks). Combining γ (whether fusion is worthwhile) and Δ_{map} (whether transfer is possible), Fig. 9 provides a $\gamma \times \Delta$ a priori decision diagram: low γ low $\Delta \rightarrow$ use simple additive model and transfer is feasible; high $\Delta \rightarrow$ per-cohort recalibration needed; only when γ is significantly large is complex fusion worthwhile.

6.10. Complexity and Extensibility

The simple additive functional model training consists of FPCA ($\mathcal{O}(n|\Lambda|K)$, n pixel count, $K=15$) plus neighborhood-mean filtering ($\mathcal{O}(nw^2K)$, w window width) and one ≤ 480 -parameter linear readout, completing in seconds on a single CPU; inference is a single matrix multiplication, inherently suitable for edge/real-time deployment. Its parameter count is $168 \sim 859 \times$ lower than the deep baselines in this paper (Table 7), implying substantially lower memory and energy consumption. Deep spatial–spectral model convolution forward pass is $\mathcal{O}(nw^2Kc^2d)$ (c channel count, d layer count), requiring GPU training. Eq. (2) joint estimator via spectral FFT and sparse graph-Laplacian descent has per-step $\mathcal{O}(n|\Lambda| \log |\Lambda| + n \cdot \text{nnz}(L))$, linear in pixel count and near-linear in band count. It should be noted: Eq. (2)'s spatial regularization joint estimator serves only as *theoretical complexity* analysis in this paper; the main-text “joint” model adds low-order cross terms to additive features (Section 5.5); wall-clock and convergence curves for Eq. (2) at scale are left as future work. Combined with Fig. 5 accuracy-cost frontier: for tasks with small $\hat{\gamma}$, the simple additive model sits on the frontier in both “accuracy/parameters” and “accuracy/energy consumption” dimensions, serving as the robust default choice for resource-constrained or interpretability-required deployment scenarios.

6.11. Negative Controls and Data Integrity

To rule out “spurious signal” and workflow-level leakage, we perform three negative controls on the simple additive model (leakage-safe spatial blocking CV; run on 6000 pixel subsample for cost control, so absolute F1 differs slightly from Table 7; focus on *relative* comparisons). Results are in Table 16. (i) **Label permutation**: shuffle training labels and retrain 300 times;

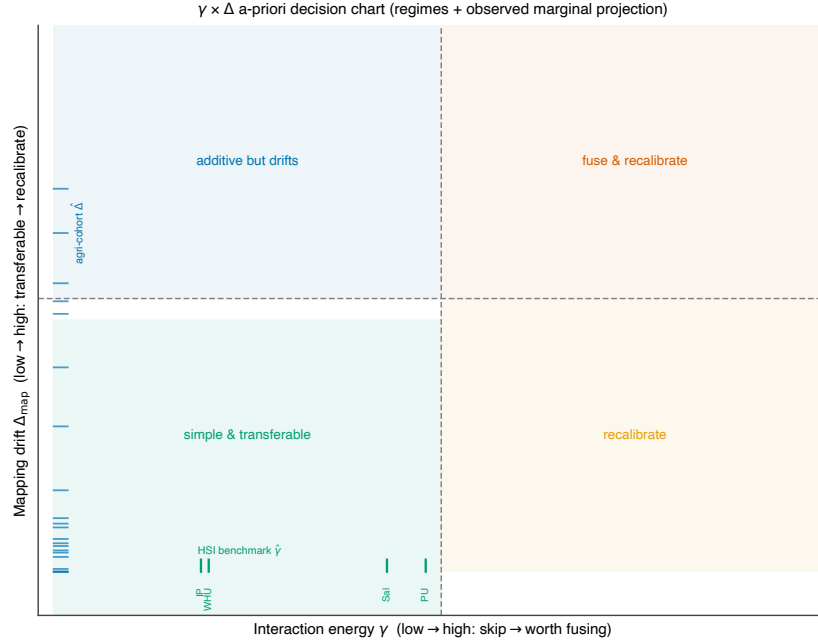


Figure 9: $\gamma \times \Delta$ a priori decision diagram (illustrative): based on interaction energy γ and mapping drift Δ_{map} for pre-deployment verdict on “whether fusion is needed” and “whether transfer is possible”; exact thresholds depend on task calibration.

real macro F1 far exceeds the permutation null distribution (mean $0.03 \sim 0.07$), with none of 300 permutations exceeding the real value ($p \leq 0.003$), ruling out “model wins by memorizing labels.” (ii) **Random features:** replace input with same-shape Gaussian noise, keeping labels; macro F1 drops to $0.01 \sim 0.07$ (near random), confirming that the model indeed uses real spatial–spectral signals rather than preprocessing artifacts. (iii) **ID probe:** use only pixel normalized coordinates as features—under leakage-safe spatial blocking macro F1 is low ($0.10 \sim 0.41$, test block coordinates unobserved during training, no generalization), while switching to random splitting significantly elevates it (Salinas $0.405 \rightarrow 0.599$, Indian Pines $0.155 \rightarrow 0.230$), *independently* corroborating that random splitting introduces spatial leakage. All three controls consistently indicate that this paper’s results are not spurious signals, and further corroborate that evaluation leakage genuinely exists.

Subsampling does not destroy spatial structure or artificially compress model gaps: To address the concern that class-quota subsampling ($\leq 8000 \sim 9000$ pixels) might weaken spatial auto-correlation and thus contradict the leakage-safe claim, we quantify spatial auto-correlation of the label field *before and after* subsampling (per-class one-vs-rest Moran’s I weighted by class frequency): all three benchmarks have full-set Moran’s $I = 0.998 \sim 1.000$, remaining $0.955 \sim 0.988$ after subsampling (minimum one-class permutation $z = 145.9$, all significantly positive), confirming that strong spatial auto-correlation

structure is fully preserved—leakage-safe spatial blocking remains necessary and effective after subsampling. Regarding the concern that “multiclass F1 = 0 (e.g., Indian Pines Corn/Grass-pasture-mowed/Oats, Salinas Vinyard-trellis) might be a subsampling artifact that artificially compresses model gaps”: these classes are *genuinely* extremely rare in the full set ($20 \sim 237$ pixels, further reduced by subsampling) or spectrally highly confused near class margins, and are *real* extremely difficult classes rather than sampling artifacts; removing them elevates both simple additive and best deep macro F1, while their *gap remains nearly unchanged* (e.g., Indian Pines $+0.056 \rightarrow +0.069$, Salinas $+0.044 \rightarrow +0.047$), indicating that zero-F1 classes pull down both models *equally* and do not artificially inflate the “simple approaches deep” conclusion.

7. Discussion

Consolidating all experiments across four benchmarks under the FPCA+neighborhood-mean featurization, four consistent findings emerge.

(1) **Second-order dominance of interaction energy.** Spatial–spectral interaction energy is overwhelmingly second-order ($\rho_2 = 0.54 \sim 0.80$), with higher-order contribution consistently small (IP and PaviaU $\gamma_{(3+)} \leq 0.010$; Salinas higher-order estimate indistinguishable from noise). This result is cross-validated by five complementary methods (Hoeffding, Sobol- R^2 , mutual information, distance covariance, Hessian norm).

Table 16

Negative controls and data integrity (simple additive model, leakage-safe spatial blocking CV, 6000 pixel subsample). Real macro F1 far exceeds label permutation null distribution; random features near random; ID probe elevated under random splitting, corroborating spatial leakage.

Dataset	Real F1	Perm. null	p	Rand. feat.	ID (spatial)	ID (random)
Indian Pines	0.470	0.032	≤ 0.003	0.023	0.155	0.230
Pavia University	0.859	0.068	≤ 0.003	0.067	0.104	0.116
Salinas	0.889	0.032	≤ 0.003	0.011	0.405	0.599

(2) Additive spatial context drives fusion gain. Low-order explicit cross terms provide negligible additional benefit (5-seed macro F1 gain ≤ 0.022 , with two datasets showing joint slightly below additive). The additional gain from deep models, which can represent higher-order interactions, varies by dataset ($+0.06 \sim +0.18$ macro F1, $K_{\text{outer}} = 5$) and is statistically significant (Nadeau–Bengio t -test, $p < 0.01$). Finding (1) theoretically explains finding (2): higher-order interaction energy upper-bounds the structural gain space for deep models, and both are restricted across all four benchmarks.

(3) Parameter efficiency. A ~ 270 – 480 parameter additive model achieves 75% $\sim$ 93% of the best deep model’s macro F1 while using $1/168 \sim 1/859$ of the parameters.

(4) Evaluation leakage. Random splitting inflates absolute accuracy by up to $+38.6$ pp ($K_{\text{outer}} = 5$) and compresses real inter-model differences.

We emphasize that finding (1) is an *empirical observation under a specific featurization and evaluation protocol*, not a claim that HSI spatial–spectral interaction is inherently weak. Different featurizations (e.g., full-band kernel interaction, deep implicit representations) may reveal stronger interaction signals. This paper’s core contribution lies in providing two a priori computable diagnostics— $\hat{\gamma}$ and $\hat{\Delta}$ —that allow researchers to make quantitative, data-specific judgments about fusion necessity and cross-scenario transferability. This aligns with the broader recognition that simple models can rival deep models on certain data types [39].

Deep advantage source: stronger spatial context, rather than non-additive interaction. A natural question is: given that low-order explicit interaction gain is near zero, why do deep models still significantly outperform the simple additive model on all benchmarks? The ablation experiment (Table 13) provides a clue: spatial context gain ($+0.062$) far exceeds the interaction term ($+0.001$), while the deep model improvement ($+0.058$) is of the same order as the spatial context gain. This indicates that deep model advantage mainly comes from *stronger additive spatial context modeling ability* (larger receptive field, layered features, end-to-end optimization), rather than from capturing non-additive spatial–spectral interaction. In other words, on this task, a deep CNN is essentially

a much stronger *spatial context extractor*, rather than a spatial–spectral interaction modeler. This explains the observation in Table 7 that attention mechanisms (SE/CBAM) provide no systematic advantage—if interaction were key, attention mechanisms should yield substantial gain; but in reality they do not.

Practical implications: (i) Before investing in complex spatial–spectral fusion networks, first estimate $\hat{\gamma}$ as a screening signal: if small and deep baseline gain is also modest, then a ~ 270 – 480 parameter additive functional model can approach the optimum, with greater explainability and lower computational cost; (ii) reported accuracy must use leakage-safe spatial splitting, otherwise absolute numbers are neither comparable nor credible; (iii) when generalizing to new instruments/locations/phenology, first compute the mapping drift certificate to decide whether recalibration is needed.

Limitations. (i) The lower-bound skeletons of the propositions belong to known theory (ANOVA decomposition, posterior drift); this paper’s novelty lies in HSI instantiation and in making them computable diagnostics. (ii) The evaluation is limited to pixel-level classification on four benchmarks. Deep baselines cover 2D/3D-CNN and spatial–spectral Transformer, but recent state-space models (SpectralMamba) and graph convolutions (GCN) are not yet included. However, the central thesis is that when $\hat{\gamma}$ is small, simple additive models approach *representationally stronger* joint models; stronger architectures constrained by small interaction energy are not expected to alter this qualitative conclusion, though this extrapolation requires further validation. Extension to continuous-target regression and larger-scale data also awaits testing. (iii) The simple-to-deep macro F1 gap, while modest ($+0.06 \sim +0.18$, $K_{\text{outer}} = 5$), is statistically significant (Nadeau–Bengio t -test, all benchmarks $p < 0.01$, Cohen’s $d = 3.3 \sim 5.6$). This paper’s claim rests on γ - and drift-informed a priori screening and weighted accuracy-cost trade-off, not on rejecting deep methods. (iv) **LLH-GBID limitation.** GBIDS is a conditional relative metric whose values depend on the indicator set, weights, baseline, and (q_B, q_W) parameters; it does not constitute a uniquely true ranking. This paper uses benefit-type linear utility, equal dimension weights, theoretical-optimum baseline, and $(q_B = q_W = 2)$ as the

primary configuration. Rankings may shift under different weight or baseline settings (sensitivity analysis in Table 8).

Threats to validity: (Population validity) This paper’s conclusions are established on pixel-level land-cover/crop classification and agricultural HSI cohort data; extrapolation to continuous regression, time-series HSI, and larger-scale scenarios requires further validation. (Construct validity) Interaction energy $\hat{\gamma}$ is estimated as the low-order spectral $\times$ spatial cross terms’ interpretable variance fraction on additive residuals; synthetic validation (Fig. 3(a)) demonstrates that it *systematically underestimates* true γ under higher-order interaction (median recovery only 27%), hence $\hat{\gamma}$ should be interpreted as a *conservative lower bound* for interaction energy rather than an unbiased estimate—“ $\hat{\gamma}$ small” alone cannot assert that interaction is zero, and must be accompanied by the independent corroboration that the deep model gain is also small. A deeper limitation: this paper’s “interaction” is operationalized as FPCA score $\times$ neighborhood-mean low-order cross terms; the original cube may contain spatial–spectral interaction through channels not captured by the current featurization (e.g., non-FPCA principal component spatial distributions, nonlinear joint statistics across bands). Precise characterization of interaction energy under richer representational parameterizations (kernel methods, deep interaction layers, full-band cross terms) remains an open problem. (Internal validity) Leakage-safe evaluation uses 6×6 spatial leave-block-out; Section 6.8 buffer zone sensitivity analysis quantifies residual leakage at block boundaries: adding 1-block buffer systematically shrinks CNN2D advantage (Indian Pines drop 78%, PaviaU drop 70%, Salinas drop 55%), indicating that part of the deep gain is inflated by boundary leakage; after buffer correction, deep models still outperform simple additive models with a modest margin (≤ 0.10), not altering the directional conclusion. (Statistical-conclusion validity) Deep model accuracy-cost comparison is elevated to 5 seeds mean $\pm$ std + Nadeau–Bengio paired t -test ($p < 0.01$, Cohen’s $d = 3.3 \sim 5.6$), establishing that the deep advantage is statistically significant. The transfer dichotomy is jointly supported by permutation test (median difference $p = 9 \times 10^{-4}$), Mann–Whitney test ($p = 2.6 \times 10^{-4}$), bootstrap 95% CI (non-overlapping between groups), and Cliff’s $\delta = 0.88$ (Section 6.6); however, the transfer certificate–gap partial correlation is based on 21 cells with limited sample size, and cross-dataset extrapolation requires further validation.

Future work: Four directions are worth pursuing: (i) re-evaluate γ under richer interaction parameterizations (full-band kernel interaction, deep implicit interaction layers) to test whether the current FPCA featurization misses important interaction structure; (ii) extend the additive $\oplus$ interaction decomposition to functional regression with continuous targets and multi-indicator joint prediction, testing whether the limited benefit of low-order explicit interactions holds across task types; (iii) establish confidence intervals and significance tests for γ and Δ_{map} estimates, allowing a priori screening with statistical guarantees; (iv) combine the mapping drift certificate with cohort-adaptive recalibration to form an “a priori verdict $\rightarrow$ conditional recalibration” deployment pipeline, and quantify its benefit in reducing labeling and retraining costs in real cross-year operational scenarios.

Broader impact: This paper’s methodology has implications beyond HSI: any “continuous signal $\times$ discrete structure” multimodal fusion problem (e.g., time series $\times$ sensor, spectral $\times$ morphology, text $\times$ graph structure) can apply the “additive $\oplus$ interaction” decomposition and interaction energy γ a priori diagnosis, thus providing a computable, quantitative answer to “whether fusion is worthwhile” before investing in complex joint architectures. At the evaluation layer, this paper’s quantification of spatial leakage serves as a reminder: in any domain where training/test samples exhibit structural correlation (spatial, temporal, hierarchical, phylogenetic), random splitting may systematically inflate generalization, and structure-aware cross-validation should be adopted [30, 29]. These two points have universal value for trustworthy, reproducible, resource-efficient machine learning.

8. Conclusion

We applied functional ANOVA decomposition to HSI spatial–spectral fusion, and further decomposed the total interaction energy γ into second-order and higher-order components. Theory indicates: the larger the second-order fraction $\rho_2 = \gamma_{(2)}/\gamma$, the closer shallow additive-bilinear models approach the theoretical optimum, with the deep model structural gain space upper-bounded by $\gamma_{(3+)}$. Four-benchmark leakage-safe empirical results consistently validate, *under this study’s FPCA+neighborhood-mean featurization*: (1) second-order interaction overwhelmingly dominates ($\rho_2 = 0.54 \sim 0.80$, higher-order contribution small or indistinguishable from estimation noise), cross-validated by five complementary methods; (2) additive spatial context contributes the majority of fusion gain, with low-order explicit spatial–spectral interaction providing extremely small additional benefit (≤ 0.022 macro F1); (3) a ~ 270 – 480 parameter additive model achieves 75% $\sim$ 93% of deep baseline performance at a fraction of the parameters; (4) random

splitting inflates absolute accuracy by up to +38.6 pp ($K_{\text{outer}} = 5$). Two a priori computable diagnostics ($\hat{\gamma}$ with order decomposition and $\hat{\Delta}$) provide actionable screening tools for assessing fusion necessity and cross-scenario transferability on one’s own data.

CRediT authorship contribution statement

Panlin Li: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft. **Yuchang Li:** Software, Validation, Data curation, Writing – review & editing. **Hua Huang:** Conceptualization, Supervision, Funding acquisition, Project administration, Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data & Code availability

The four public benchmarks used in the primary experiments—Indian Pines, Pavia University, Salinas—were obtained from the EHU hyperspectral remote sensing scene repository (http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes); WHU-Hi-LongKou was obtained from the Wuhan University RSDIA group public dataset. The 7 agricultural spectral datasets used for cross-cohort transfer analysis (IDs 001/006/012/013/018/019/029) are all public except 001 (authors’ internal collection, contact corresponding author); exact sources (Mendele/Figshare/USDA Ag Data Commons etc.) are in the main-text data card (Table 2) footnotes. The complete source code for all experiments, figures, tables, and diagnostic estimators follows open-source principles; to support reproducibility during peer review, we have prepared *anonymous* code and preprocessing data packages, available upon request during the editorial or review process, and will be publicly released in a public repository upon paper acceptance (with SHA-256 checksums and environment-locked files).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used Anthropic Claude Opus 4.8, primarily for

language polishing and expression refinement. After using this tool, the authors reviewed and edited all content, and take full responsibility for the content, data, and conclusions of this paper.

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A. Proofs of Propositions

This appendix provides complete proofs for Propositions 1–4 in Section 4. Notation follows Sections 3–4.

A.1. Proposition 1 (Joint identifiability)

PROOF. Let the joint distribution $P_{s,a}$ define the additive function closed subspace

$$\mathcal{H}_{\text{add}} = \overline{\{g_1(s) + g_2(a) : g_1 \in L^2(P_s), g_2 \in L^2(P_a)\}} \subseteq L^2(P_{s,a}). \quad (10)$$

Step 1 (Interaction residual uniqueness): From the Hilbert space projection theorem, the orthogonal projection $\Pi_{\text{add}} f^*$ of f^* onto the closed subspace $\mathcal{H}_{\text{add}}$ exists and is unique, hence

$$C := f^* - \Pi_{\text{add}} f^* \quad \text{is uniquely determined.} \quad (11)$$

This conclusion *does not require* s, a independence—in HSI, s, a are derived from the same FPCA field and are strongly correlated, but projection geometry depends only on the joint measure $P_{s,a}$.

Step 2 (Additive splitting): If $\overline{L^2(s)} \cap \overline{L^2(a)} = \{\text{constants}\}$ (trivial intersection), then the centering constraints $\mathbb{E}[\phi_s] = \mathbb{E}[\phi_a] = 0$ uniquely determine the splitting $\Pi_{\text{add}} f^* = \phi_s + \phi_a$ (dependent-input functional ANOVA, see Hooker [24]); otherwise ϕ_s, ϕ_a differ by at most a common direction, while C and γ are unaffected. $\square$

A.2. Proposition 2 (Fusion gain $\asymp$ interaction energy)

PROOF. *Step 1 (Population lower bound):* For any $h \in \mathcal{H}_{\text{add}}$, from $f^* = (\phi_s + \phi_a) + C$ and projection orthogonality,

$$\inf_{h \in \mathcal{H}_{\text{add}}} \mathbb{E}[(f^* - h)^2] = \mathbb{E}[(C - \Pi_{\text{add}} C)^2] = \gamma, \quad (12)$$

i.e., concatenation (additive late fusion) population excess risk is exactly γ .

Step 2 (Upper bound and sandwich): The joint class contains f^* , so its population excess risk is 0, and under

finite samples $\leq \varepsilon_{\text{joint}}(n)$. Hence

$$\text{Gap}_{\text{pop}} = \gamma, \quad \mathbb{E} \text{Gap} \in [\gamma - \varepsilon_{\text{joint}}(n), \gamma + \varepsilon_{\text{add}}(n)]. \quad (13)$$

In particular $\gamma = 0 \Rightarrow \text{Gap}_{\text{pop}} = 0$ (concatenation is population-optimal).

Step 3 (Functional rate): When $C \in W^{\bar{s},2}$ (spectral-spatial tensor Sobolev), the target covariance is trace-class with eigenvalues $\asymp j^{-2r}$, the regularized least-squares interaction estimation error satisfies

$$\varepsilon_{\text{joint}}(n) \asymp n^{-\frac{2\bar{s}}{2\bar{s}+d_{\text{eff}}}}, \quad (14)$$

where d_{eff} is the effective dimension determined by covariance spectral decay; Eq. (2) spatial regularization reduces d_{eff} . $\square$

A.3. Proposition 4 (Cross-cohort transfer gap lower bound and certificate)

PROOF. *Step 1 (Two-point construction):* Take $\beta^{(0)}$ as the Σ_{c^*} -closest point in $\text{conv}\{\beta_c\}$, and $\beta^{(1)} = \beta^{(0)} + \Delta_{\text{map}}v$, where v is the unit residual direction attaining the infimum ($\|v\|_{\Sigma_{c^*}} = 1$). Construct two worlds P_0, P_1 : the source cohort distribution, source mapping, and target covariate law $P_{c^*}^X$ are identical; only the target conditional mean differs (taking $\beta^{(b)}$).

Step 2 (Minimax lower bound): A zero-sample estimator $\hat{f}$ has no access to target labels, and the two worlds produce the *same observable random variables*; from the inequality $(a - u)^2 + (a - w)^2 \geq \frac{1}{2}(u - w)^2$, taking expectation pointwise,

$$\max_{b \in \{0,1\}} \mathbb{E} \text{ER}_{c^*}(\hat{f}; b) \geq \frac{1}{4} \mathbb{E}_{x \sim P_{c^*}^X}[(\beta^{(1)} - \beta^{(0)}, x)^2] = \frac{1}{4} \Delta_{\text{map}}^2. \quad (15)$$

This is a minimax statement over target classes (not a pointwise claim for fixed β_{c^*}).

Step 3 (Certificate consistency): Within-cohort least squares $\hat{\beta}_c \rightarrow_p \beta_c$ (spectral gap ensures the leading direction is unique), and the pairwise Σ -metric $\hat{\Delta}$ is a monotone consistent estimator of Δ_{map} , hence computable a priori. $\square$

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